

Fair Conformal Prediction for Individuals and Subgroups in Natural and Adversarial Settings: Theoretical Guarantees and Impossibility Results

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Abstract

Conformal prediction offers a principled, model-agnostic framework for constructing prediction sets with formal coverage guarantees under the assumption of exchangeability. In many practical applications, however, coverage alone is insufficient, and additional properties of the resulting prediction sets are often required. For instance, recent work has explored how to ensure CP fair treatment of both individuals and population subgroups and robustness to adversarial perturbations. In this paper, we study whether it is possible to construct conformal prediction sets that preserve coverage guarantees while simultaneously ensuring fair treatment of individuals, in the sense of counterfactual fairness by means of Equal Set Size, and of population subgroups, in the sense of Equalized Coverage and Equalized Average Set Size, in both natural and adversarial settings. In the adversarial setting, we consider an attacker whose goal is to induce unfair behavior toward either individual subjects or population subgroups, and we develop the first attack and defense methods for this scenario. We analyze both the case in which the sensitive attribute is available at test time and the more realistic setting in which it is not. For the natural setting, we propose conformal prediction methods that are provably efficient and satisfy fairness guarantees, and we also establish corresponding impossibility results. For the adversarial setting, we show that, under a realistic threat model in which the adversarial strategy is explicitly specified, the guarantees achieved in the natural setting can be recovered. Finally, experiments on real-world classification datasets involving fairness-sensitive tasks demonstrate the effectiveness and practical relevance of our approach, while also highlighting the limitations of existing methods.

Keywords: Fair Conformal Prediction, Individual Fairness, Counterfactual Fairness, Equalized Set Size, Subgroups Fairness, Equalized Coverage, Equalized Average Set Size, Adversarial Robustness, Distribution-Free Coverage Guarantees, Fairness under Adversarial Perturbations, Impossibility Results.

1. Introduction

In many modern applications, Artificial Intelligence (AI) systems, particularly those based on Machine Learning (ML), are expected not only to produce accurate predictions but also to provide reliable and transparent quantification of their uncertainty. Uncertainty quantification is crucial for informed decision-making, especially in high-stakes domains where erroneous or overconfident predictions may have serious consequences, such as healthcare (Zou et al., 2023), finance (Dixon et al., 2020), transportation (Wang et al., 2023b), and logistics (Akbari and Do, 2021). Among the available approaches, Conformal Prediction (CP) has emerged as a principled, model-agnostic framework for constructing prediction sets with formal coverage guarantees under mild assumptions (Vovk et al., 2005).

However, coverage alone is often insufficient to fully characterize the reliability of a predictive system. This limitation is also reflected in emerging regulatory frameworks, such as the Data Act¹ and the proposed AI Act² in Europe, as well as HIPAA³ in the United States, which emphasize not only reliability but also broader requirements, including fairness and robustness in AI systems. Fairness concerns whether a model treats individuals and groups equitably (Mehrabi et al., 2021a; Barocas et al., 2023; Hardt et al., 2016; Dwork et al., 2012; Kusner et al., 2017), whereas robustness concerns the stability of predictions under small, potentially adversarial, perturbations of the input (Biggio et al., 2013; Biggio and Roli, 2018; Vaccari et al., 2022). A system that provides valid uncertainty estimates but behaves unfairly or remains vulnerable to manipulation may still be unsuitable for deployment in sensitive applications.

Fairness, in particular, has been studied extensively in the ML literature from multiple perspectives. Group-level notions aim to ensure that different subgroups, identified by sensitive attributes such as gender, ethnicity, or age, are treated in a statistically comparable manner, for example through criteria such as *Demographic Parity* or *Equal Opportunity* (Hardt et al., 2016; Agarwal et al., 2019). By contrast, individual-level notions focus on the treatment of single instances, requiring predictions to remain stable under hypothetical changes in sensitive attributes, as formalized by *Counterfactual Fairness* (Kusner et al., 2017). These two perspectives capture complementary aspects of fairness at the population and individual levels. At the same time, it is well known that different fairness criteria are generally incompatible, as established by several impossibility results in the literature (Kleinberg et al., 2016; Chouldechova, 2017). For example, *Demographic Parity* and *Equal Opportunity* cannot, in general, be satisfied simultaneously. This suggests that any attempt to combine multiple fairness notions must carefully account for these fundamental limitations.

Within the CP framework, fairness can be naturally formulated by requiring prediction sets to attain comparable coverage across subgroups defined by sensitive attributes. This idea has been explored in group-conditional CP (Romano et al., 2020b), class-conditional CP (Ding et al., 2024), and, more recently, in approaches that provide formal fairness guarantees through quantile regression (Liu et al., 2022). This principle, often referred to as *Equalized Coverage* (Romano et al., 2020a) or *Equalized Average Set Size* (Cresswell et al., 2025), aims to prevent prediction sets from systematically favoring or disadvantag-

1. <https://digital-strategy.ec.europa.eu/en/policies/data-act>

2. <https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>

3. <https://www.hhs.gov/hipaa/index.html>

ing particular subgroups. In practice, it can be enforced by calibrating quantile thresholds separately for each group, for example through Mondrian CP (Vovk et al., 2003). In this sense, *Equalized Coverage* or *Equalized Average Set Size* may be viewed as a natural analogue of *Demographic Parity* in the context of set-valued predictions. At the same time, individual-level fairness can be expressed through the invariance of prediction sets under changes in sensitive attributes, as formalized in Counterfactually Fair CP (Guldogan et al., 2025). Under this notion, prediction sets are required to remain identical for factual and counterfactual instances that differ only in the sensitive attribute. In this setting, fairness is reflected in *Equalized Set Size*, in the sense defined by Cresswell et al. (2025), at the individual level between factual and counterfactual instances, while coverage is guaranteed only in a marginal sense.

The goal of this paper is to construct CP sets that combine individual-level and group-level fairness in a practical and theoretically grounded manner (Section 4). Specifically, we seek to construct prediction sets that enforce *Equalized Set Size* at the individual level across counterfactual instances, in the sense of counterfactual fairness (Guldogan et al., 2025), while simultaneously satisfying *Equalized Coverage* and *Equalized Average Set Size* across subgroups, as introduced in (Romano et al., 2020a; Cresswell et al., 2025). The resulting prediction sets therefore enforce *Equalized Set Size* across subgroups while also satisfying *Equalized Coverage* and *Equalized Average Set Size*. We emphasize that this result can be achieved only when the sensitive attribute is available at test time; when it is not, exact equality in both average set size and coverage becomes unattainable. We present the latter as an impossibility result, adding to the large body of work on fairness impossibility theorems (Kleinberg et al., 2016; Chouldechova, 2017), such as results showing the impossibility of simultaneously satisfying *Demographic Parity* and *Equal Opportunity*. In the more realistic setting in which the sensitive attribute is unavailable at test time (Dwork et al., 2018), we adopt a conservative strategy that selects the largest among the group-specific quantiles obtained through Mondrian calibration. This choice preserves *Equalized Set Size* while guaranteeing *Equalized Coverage* at the desired level across groups, at the cost of increased marginal coverage due to over-coverage in some groups.

In adversarial settings (Section 5), both fairness and coverage guarantees may be compromised by an adversary who deliberately perturbs the input to induce unfair behavior, either against individual subjects or across entire subgroups (Biggio et al., 2013; Biggio and Roli, 2018; Vaccari et al., 2022). To study and mitigate this risk, we design a gradient-based fairness attack that simultaneously targets individual-level *Counterfactual Fairness*, through violations of *Equalized Set Size*, and group-level fairness, through violations of *Equalized Coverage* or *Equalized Average Set Size*, thereby explicitly amplifying unfairness in the outputs of conformal prediction sets across different realizations of the sensitive attribute. To defend against such attacks, we extend the conformal prediction framework developed for the natural setting by incorporating adversarial robustness techniques. In particular, we build on our previous work on Provably Efficient and Robust CP (Carlevaro et al., 2025), which provides coverage guarantees even under carefully crafted input perturbations. Our approach applies both when the sensitive attribute is available at test time and in the more realistic scenario in which it is unavailable. To the best of our knowledge, this is the first framework to jointly combine individual-level *Counterfactual Fairness*, enforced through *Equalized Set Size*, group-level fairness, enforced through *Equalized Coverage* and *Equalized*

Average Set Size, and adversarial robustness within a unified conformal prediction setting, while offering both theoretical guarantees and practical applicability in realistic scenarios. Finally, we validate the proposed framework through empirical evaluations on real-world classification datasets involving fairness-sensitive tasks (Section 6). Our experiments show that the constructed conformal prediction sets effectively enforce individual-level *Counterfactual Fairness*, through *Equalized Set Size*, and group-level fairness, through *Equalized Coverage* and *Equalized Average Set Size*, in the natural setting, while preserving a meaningful degree of efficiency. We also assess the robustness of the approach under adversarial perturbations, showing that fairness and coverage guarantees can be maintained even in the presence of attacks specifically designed to amplify disparities across sensitive attributes. All code and experiments are publicly available in a GitHub repository⁴, thereby ensuring reproducibility and supporting further research on fairness and robustness in conformal prediction.

2. Related Work

In this section, we review the literature on fairness and adversarial robustness in CP. To the best of our knowledge, no existing work addresses, within the standard setting, both individual-level *Counterfactual Fairness*, enforced through *Equalized Set Size*, and group-level fairness, enforced through *Equalized Coverage* and *Equalized Average Set Size*. Moreover, the CP literature does not, to our knowledge, study an adversarial setting in which carefully designed perturbations explicitly aim to exacerbate unfairness at both the individual and group levels, together with a corresponding defense mechanism supported by theoretical guarantees. For these reasons, we discuss these two research directions separately and stress that their integration cannot be obtained through a simple combination of existing methods.

Fairness in Conformal Prediction. Fairness in CP has been studied primarily through group-level guarantees, with the objective of ensuring that prediction sets attain comparable coverage across subgroups defined by sensitive attributes. A seminal contribution in this direction is due to Romano et al. (2020a), who introduced the notion of *Equalized Coverage*, which requires prediction sets to achieve the same coverage level across predefined groups. This principle can be implemented through group-wise calibration procedures, such as Mondrian CP (Vovk et al., 2003), and has inspired a broad line of subsequent work. Similarly, Cresswell et al. (2025) introduced the notion of *Equalized Average Set Size*, which strengthens the coverage requirement by additionally imposing that prediction sets have the same average size across predefined groups.

Several extensions have investigated alternative statistical notions of group-level fairness within the CP framework. For example, Liu et al. (2022) incorporate constraints inspired by *Demographic Parity*, while Wang et al. (2023a) introduce *Equal Opportunity of Coverage*, which conditions coverage guarantees on the outcome variable. More recent contributions propose adaptive or optimization-based strategies to balance fairness and efficiency, including (Zhou and Sesia, 2024; Garcia-Galindo et al., 2025; Sun et al., 2025; Gao et al., 2025). Despite their differences, these approaches share a common perspective: fairness is

4. https://github.com/AlbiCarle/FAIR_and_ROBUST_CP

enforced at the group level by controlling coverage, either marginally or conditionally on group membership.

However, recent evidence suggests that equalizing coverage alone may be insufficient to ensure fair outcomes. In particular, [Cresswell et al. \(2025\)](#) show that enforcing *Equalized Coverage* may even amplify disparities in downstream decisions, and argue that properties such as prediction set size, namely *Equalized Set Size*, or singleton frequency may be more directly aligned with fairness objectives.

This observation motivates the study of individual-level notions of fairness. From this perspective, requiring prediction sets to remain invariant under changes in sensitive attributes naturally leads to enforcing *Equalized Set Size* across counterfactual instances. Building on this idea, [Guldogan et al. \(2025\)](#) propose symmetrizing the conformity score across different realizations of the sensitive attribute, for instance by means of aggregation operators such as the mean, maximum, or minimum. As a result, the corresponding quantile is shared across subgroups, yielding prediction sets that coincide for factual and counterfactual instances.

However, this construction provides coverage guarantees only at the marginal level and does not prevent disparities across groups. Conversely, methods that enforce group-level guarantees through group-specific calibration achieve coverage control within each group, but generally induce different quantiles and, consequently, different prediction set sizes across subgroups ([Romano et al., 2020a](#)).

Existing methods largely focus on either group-level fairness or individual-level fairness in isolation, and do not provide a unified framework that jointly addresses both notions within CP ([Romano et al., 2020b](#); [Guldogan et al., 2025](#); [Wang et al., 2023a](#); [Zhou and Sesia, 2024](#); [Berk et al., 2024](#); [Gibbs et al., 2025](#); [Garcia-Galindo et al., 2025](#)). In addition, most of these approaches assume that the sensitive attribute is available at test time, an assumption that may be unrealistic in many practical applications ([Dwork et al., 2012](#)).

Recent work has extended fairness-aware CP to more complex settings, including large language models and recommendation systems ([Fayyazi et al., 2025](#); [Liu et al., 2026](#)), but these approaches still adopt a single notion of fairness and do not address the interaction between group-level and individual-level guarantees.

In this work (Section 4), we address this gap by developing a CP framework that explicitly combines individual-level *Counterfactual Fairness*, enforced through *Equalized Set Size*, with group-level fairness, enforced through *Equalized Coverage* and *Equalized Average Set Size*. We consider both the standard setting and the more realistic scenario in which the sensitive attribute is unavailable at test time. For the latter setting, we further establish an impossibility result showing that *Equalized Coverage* and *Equalized Average Set Size* cannot, in general, be achieved simultaneously, thus adding to the broad literature on fairness impossibility theorems ([Kleinberg et al., 2016](#); [Chouldechova, 2017](#)).

Adversarial Robustness in Conformal Prediction. The robustness of CP under adversarial perturbations has recently received growing attention. Standard CP guarantees rely on exchangeability and are known to deteriorate under distribution shifts, including adversarial attacks. To address this limitation, several methods have been proposed to provide robustness guarantees against worst-case perturbations.

A first line of work relies on randomized smoothing techniques, such as RSCP+ ([Gendler et al., 2021](#)) and subsequent extensions ([Yan et al., 2024](#)), which achieve robustness by

averaging predictions over randomized perturbations of the input. Alternative approaches are based on verification procedures, such as VRCP-I and VRCP-C (Jeary et al., 2024), which provide certified guarantees by explicitly bounding the effect of adversarial perturbations under a specified threat model. More recently, efficient robust CP methods have been proposed to improve the trade-off between robustness and statistical efficiency (Zargarbashi et al., 2024, 2025; Carlevaro et al., 2025).

Despite these advances, existing approaches focus primarily on preserving coverage guarantees under adversarial perturbations and do not account for fairness considerations. In particular, robustness is typically studied with respect to marginal or group-conditional coverage, without analyzing how adversarial perturbations may affect the fairness properties of prediction sets.

In parallel, a growing body of work in adversarial machine learning has shown that fairness itself can become a target of adversarial manipulation (Xu et al., 2021; Wang and Huang, 2019; Zhang et al., 2018). Both evasion and poisoning attacks have been proposed to amplify disparities across sensitive groups or to undermine fairness guarantees, even when models are explicitly designed to be fair (Mehrabi et al., 2021a). These results highlight that fairness and robustness are deeply intertwined, and that enforcing fairness in the natural setting does not imply resilience in the adversarial one.

However, the interaction between adversarial robustness and fairness in CP remains largely unexplored. To the best of our knowledge, no existing work studies adversarial attacks that explicitly target fairness properties in CP, nor proposes defenses that jointly guarantee coverage and fairness under adversarial perturbations.

In this work (Section 5), we take a first step in this direction by introducing a gradient-based attack that explicitly targets fairness criteria in CP. Specifically, we design perturbations that amplify discrepancies either across groups, thereby affecting *Equalized Coverage* and *Equalized Average Set Size*, or across counterfactual realizations of the same input, thereby affecting *Equalized Set Size*. We further show how our previous work on provable efficiency robustness in CP (Carlevaro et al., 2025) can be adapted to restore coverage guarantees under adversarial perturbations while preserving fairness properties. The resulting framework jointly addresses fairness and robustness in conformal prediction, a combination that, to the best of our knowledge, has not been explored by existing methods.

3. Background Concepts

Before presenting the theoretical contributions of this work, we introduce the fundamentals of multiclass classification, Mondrian CP, fairness, and adversarial machine learning. This section also establishes the notation and key concepts used throughout the paper.

3.1. Multiclass Classification

In this work, we focus on multiclass classification⁵ (Shalev-Shwartz and Ben-David, 2014; Bishop and Bishop, 2023). Let $\mathcal{X} \subseteq \mathbb{R}^d$, $d \in \mathbb{N}, d \geq 1$, denote the input space and let $\mathcal{Y} = \{1, \dots, m\}$, $m \in \mathbb{N}, m > 1$, be a finite output space. A *classifier* is a function

5. The results can also be extended to regression, but we restrict our attention to classification due to space constraints.

$f \in \mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \mathcal{Y}\}$ that assigns a label $\hat{y} \in \mathcal{Y}$ to an input sample $\mathbf{x} \in \mathcal{X}$. Typically, this prediction is determined by the model’s internal scores, such as estimated probabilities or logits, so that

$$\hat{y} = f(\mathbf{x}) = \arg \max_{y \in \mathcal{Y}} \mathbf{f}(\mathbf{x})_y, \quad (1)$$

where $\mathbf{f}(\mathbf{x})_y \in \mathbb{R}$ denotes the y -th component of the model’s internal score vector, with $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^m$. When probability estimates are required, they can be obtained by applying the softmax function $\mathbb{f}(\mathbf{x}) = \text{softmax}(\mathbf{f}(\mathbf{x}))$.

3.2. Mondrian Conformal Prediction

As discussed above, fairness in CP can be addressed through group-conditional CP, in particular through Mondrian CP, which we briefly review in this section.

For a classification task, we follow the inductive CP procedure (Shafer and Vovk, 2008). Given a labeled dataset of exchangeable examples $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, i.e., examples whose joint probability distribution is invariant under permutations, the data are partitioned into a proper training set $\mathcal{D}_{\mathcal{T}}$ and a calibration set $\mathcal{D}_{\mathcal{C}}$. The calibration set contains unseen data that are not used during training, thereby enabling an unbiased estimation of predictive uncertainty. Uncertainty is quantified through a nonconformity score function $s : \mathcal{F} \times \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ where larger values correspond to lower conformity. Common choices of nonconformity scores for classification include the Least Ambiguous Classifier (LAC) (Sadinle et al., 2019), the Adaptive Prediction Set (APS) (Romano et al., 2020b), and its regularized variant, RAPS (Angelopoulos et al., 2020). These scoring rules differ in how they trade off coverage and efficiency, but they do not affect the validity of CP. In some applications, one may require prediction sets that attain equal coverage across specific subsets of the data. Let $\mathcal{G}$ be a finite set of group labels, and let $g : \mathcal{X} \rightarrow \mathcal{G}$ be a function that assigns each input $\mathbf{x} \in \mathcal{X}$ to its corresponding group $g(\mathbf{x}) \in \mathcal{G}$. In Mondrian CP (Vovk et al., 2003), the calibration scores are stratified according to the group assignment, yielding group-specific score distributions $\mathbf{s}^{(a)} = [s(f, \mathbf{x}, y) : (\mathbf{x}, y) \in \mathcal{D}_{\mathcal{C}}, g(\mathbf{x}) = a]$. For a given confidence level $1 - \alpha \in (0, 1)$, a separate empirical quantile is computed for each group:

$$Q_{1-\alpha}^{(a)}(\mathbf{s}^{(a)}) = \min \left\{ q \in \mathbb{R} : \frac{1}{n^{(a)}} \sum_{i=1}^{n^{(a)}} [s_i^{(a)} \leq q] \geq 1 - \alpha \right\}, \quad (2)$$

where $n^{(a)}$ denotes the number of examples from group a in the calibration set, and where we use the Iverson bracket notation. For notational simplicity, we omit the explicit dependence of the quantiles $Q_{1-\alpha}^{(a)}$ on the underlying calibration scores when clear from the context. Then, for a test sample $(\mathbf{x}_{n+1}, y_{n+1})$, Mondrian CP constructs the conformal prediction set using the quantile associated with its group:

$$\mathcal{C}(\mathbf{x}_{n+1}) = \left\{ y \in \mathcal{Y} : s(f, \mathbf{x}_{n+1}, y) \leq Q_{1-\alpha}^{(g(\mathbf{x}_{n+1}))} \right\}. \quad (3)$$

Under the exchangeability assumption, this construction guarantees group-conditional coverage, namely

$$\mathbb{P} \{ y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1}) : g(\mathbf{x}_{n+1}) = a \} \geq 1 - \alpha, \quad \forall a \in \mathcal{G}. \quad (4)$$

The statistical efficiency of the resulting prediction sets, typically measured by their size $|\mathcal{C}(\mathbf{x})|$, depends on the choice of the nonconformity score. In the Mondrian CP setting, efficiency is naturally evaluated separately for each group, since the conformal threshold is calibrated independently within each group.

3.3. Fairness in Conformal Prediction

In this section, we formalize several notions of fairness that are relevant to CP, at both the group and individual levels⁶.

We consider a setting in which each input $\mathbf{x} \in \mathcal{X}$ is associated with a sensitive attribute taking values in a binary set⁷ $\mathcal{G} = \{a, \bar{a}\}$. In line with the fairness setting, we assume that $g(\mathbf{x})$ is always available for the calibration data, whereas at test time it may or may not be available (Dwork et al., 2018); that is, for a new input $\mathbf{x}_{n+1}$, the sensitive attribute may be either known or unknown.

Group-level fairness. Group-level fairness aims to ensure that prediction sets provide comparable conformity guarantees across subgroups defined by the sensitive attribute (Romano et al., 2020a; Liu et al., 2022; Wang et al., 2023a; Zhou and Sesia, 2024; Cresswell et al., 2025). Although several notions of group-level fairness have been proposed, in this work we focus on two of the most prominent ones, namely *Equalized Coverage* and *Equalized Average Set Size*.

Definition 1 (Equalized Coverage) *The conformal predictor $\mathcal{C}$ satisfies Equalized Coverage at confidence level $1 - \alpha$ if*

$$\mathbb{P} \{y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1}) : g(\mathbf{x}_{n+1}) = a\} \geq 1 - \alpha, \quad \forall a \in \mathcal{G}. \quad (5)$$

This notion ensures that each subgroup receives the same target coverage guarantee, even though the realized coverage levels need not be exactly identical across groups. It can be viewed as the analogue of *Demographic Parity* for set-valued predictions, where fairness is expressed in terms of coverage guarantees rather than point decisions.

Definition 2 (Equalized Average Set Size) *A conformal predictor $\mathcal{C}$ is said to satisfy Equalized Average Set Size if*

$$\mathbb{E} [|\mathcal{C}(\mathbf{x}_{n+1})| : g(\mathbf{x}_{n+1}) = a] = \mathbb{E} [|\mathcal{C}(\mathbf{x}_{n+1})| : g(\mathbf{x}_{n+1}) = \bar{a}]. \quad (6)$$

This notion complements *Equalized Coverage* by addressing the informativeness of prediction sets across groups. Whereas *Equalized Coverage* ensures that all groups achieve the same level of coverage, *Equalized Average Set Size* requires that no group systematically receives larger, and therefore less informative, prediction sets than another.

6. The results of this work can be extended to other notions of fairness; however, due to space constraints, we focus on a selected subset.

7. For simplicity, and due to space constraints, we restrict our analysis to the binary case. Nevertheless, the framework can be extended, with additional technical work, to multivalued or continuous sensitive attributes.

Individual-level fairness. Individual-level fairness in CP concerns the behavior of prediction sets under counterfactual changes in the sensitive attribute (Guldogan et al., 2025). To formalize this notion, let us define a counterfactual operator $c : \mathcal{X} \rightarrow \mathcal{X}$ that maps each instance to its counterfactual counterpart (Kusner et al., 2017), for example by changing the value of the sensitive attribute. Formally, for any $\mathbf{x} \in \mathcal{X}$, we write $c(\mathbf{x}) = \bar{\mathbf{x}}$, where $\bar{\mathbf{x}}$ denotes the counterfactual version of $\mathbf{x}$. Based on this, we introduce the following notion of individual-level fairness in CP, which captures invariance of prediction set size under counterfactual interventions.

Definition 3 (Equalized Set Size) *A conformal predictor $\mathcal{C}$ satisfies Equalized Set Size at the individual level if*

$$|\mathcal{C}(\mathbf{x})| = |\mathcal{C}(\bar{\mathbf{x}})|, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (7)$$

This notion enforces *Counterfactual Fairness* (Kusner et al., 2017) in the sense formalized in (Guldogan et al., 2025, Definition 3.2). In particular, Guldogan et al. (2025) extend *Counterfactual Fairness* to prediction sets by requiring that, for any individual, the probability that a label is included in the prediction set remains invariant under counterfactual changes in the sensitive attribute. Formally,

$$\mathbb{P}\{y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1})\} = \mathbb{P}\{y_{n+1} \in \mathcal{C}(\bar{\mathbf{x}}_{n+1})\} \geq 1 - \alpha, \quad (8)$$

marginally (Guldogan et al., 2025, Corollary 4.2).

In practice, this type of invariance can be achieved through constructions, for example based on mean, maximum, or minimum aggregation, that enforce symmetry between factual and counterfactual instances at the score level while preserving exchangeability (Guldogan et al., 2025, Theorem 4.1). As a result, prediction sets are constructed symmetrically across counterfactual pairs, thereby inducing pointwise counterfactual invariance of the resulting sets and, consequently, *Equalized Set Size*.

3.4. Adversarial Attacks

Adversarial machine learning studies the vulnerability of predictive models to carefully crafted perturbations of the input (Biggio and Roli, 2018; Huang et al., 2020). In this work, we focus on evasion attacks, in which an adversary seeks to modify an input $\mathbf{x} \in \mathcal{X}$ into a perturbed version $\tilde{\mathbf{x}}$ in order to alter the behavior of the predictor.

We consider perturbations constrained to lie in a set $\mathcal{P}(\mathbf{x}) \subseteq \mathcal{X}$, typically defined as an ℓ_p -ball of radius $\varepsilon > 0$:

$$\mathcal{P}(\mathbf{x}) = \{\mathbf{x}' \in \mathcal{X} : \|\mathbf{x}' - \mathbf{x}\|_p \leq \varepsilon\}. \quad (9)$$

Classical adversarial attacks are commonly formulated as constrained optimization problems of the form

$$\tilde{\mathbf{x}} = \arg \min_{\mathbf{x}' \in \mathcal{P}(\mathbf{x})} \mathcal{O}(\mathbf{x}'), \quad (10)$$

where $\mathcal{O}$ denotes a generic attack objective, typically designed to reduce the model’s confidence in the correct prediction (Biggio and Roli, 2018; Huang et al., 2020), for example through a classification loss or a margin-based surrogate.

Several families of evasion attacks have been proposed in the literature, including gradient-based methods such as FGSM, PGD, and AutoAttack (Goodfellow et al., 2014; Madry et al., 2018; Croce and Hein, 2020), optimization-based attacks (Carlini and Wagner, 2017; Chen et al., 2018; Moosavi-Dezfooli et al., 2016), and attacks tailored to physical or black-box settings (Kurakin et al., 2018; Papernot et al., 2016). Broader taxonomies of adversarial threat models and attack strategies can be found in (Pitropakis et al., 2019; Xiong and Lagerström, 2019).

While most existing work focuses on degrading predictive accuracy, in this paper we consider evasion attacks with a different objective. Specifically, we study perturbations that target the fairness properties of the predictor (Mehrabi et al., 2021b; Gittens et al., 2022). Given the group assignment function $g : \mathcal{X} \rightarrow \mathcal{G}$, the adversary seeks perturbations that amplify disparities either across groups or across counterfactual realizations of the same input under changes in the sensitive attribute.

Unlike standard attacks, the goal is therefore not misclassification, but rather the maximization of fairness violations induced by the model under bounded perturbations. The precise formulation of these fairness-oriented objectives will be introduced in Section 5.

4. Natural Setting: Group- and Individual-Level Fair CP

In this section, we develop CP that simultaneously satisfy group-level fairness, namely *Equalized Coverage* and *Equalized Average Set Size*, and individual-level fairness, namely *Equalized Set Size*, in the natural setting, that is, in the absence of adversarial perturbations. To enforce individual-level fairness, we construct nonconformity scores that are invariant under the counterfactual operator $c : \mathcal{X} \rightarrow \mathcal{X}$, where $c(\mathbf{x}) = \bar{\mathbf{x}}$. Following Guldogan et al. (2025), given a base score function $s(f, \mathbf{x}, y)$, we define the symmetrized score

$$\bar{s}(f, \mathbf{x}, y) = \text{Agg}(s(f, \mathbf{x}, y), s(f, \bar{\mathbf{x}}, y)), \quad (11)$$

where Agg is a symmetric aggregation operator, such as the mean, maximum, or minimum (Guldogan et al., 2025, Section 4.1).

We then combine this construction with Mondrian CP. More precisely, calibration scores are computed separately within each group, yielding the group-conditional score collections

$$\bar{\mathbf{s}}^{(a)} = [\bar{s}(f, \mathbf{x}, y) : (\mathbf{x}, y) \in \mathcal{D}_{\mathcal{C}}, g(\mathbf{x}) = a], \quad \forall a \in \mathcal{G}. \quad (12)$$

Let $\bar{Q}_{1-\alpha}^{(a)}$ denote the corresponding empirical quantile, computed according to Eq. (2).

When the group assignment function $g : \mathcal{X} \rightarrow \mathcal{G}$, i.e., the sensitive feature, is available at test time, the resulting CP is denoted by $\mathcal{C}^{\text{EQ}}$ and defined as

$$\mathcal{C}^{\text{EQ}}(\mathbf{x}) = \left\{ y \in \mathcal{Y} : \bar{s}(f, \mathbf{x}, y) \leq \bar{Q}_{1-\alpha}^{(g(\mathbf{x}))} \right\}. \quad (13)$$

By construction, $\mathcal{C}^{\text{EQ}}$ enforces group-level fairness through group-conditional calibration and individual-level fairness through the counterfactual invariance of the score. This yields our first main result.

Theorem 4 (Fair CP with Equalized Coverage and Equalized Set Size - Sensitive Feature Known at Test Time) *Assume that the group assignment function $g : \mathcal{X} \rightarrow \mathcal{G}$*

is available at test time. Let $\mathcal{C}^{\text{EQ}}$ be the conformal predictor defined in Eq. (13), constructed using the symmetrized score $\bar{s}$ in Eq. (11) and the group-conditional quantiles $\bar{Q}_{1-\alpha}^{(a)}$ computed according to Eq. (2), at confidence level $1 - \alpha \in (0, 1)$. If the data are exchangeable, then $\mathcal{C}^{\text{EQ}}$ satisfies both Equalized Coverage and Equalized Set Size.

Proof The proof of Theorem 4 is provided in Appendix A.1. ■

While $\mathcal{C}^{\text{EQ}}$ ensures Equalized Coverage and counterfactual invariance, it does not control the average size of the prediction sets across groups. In general, different groups may exhibit different average set sizes because of differences in the underlying score distributions. To explicitly enforce fairness also in terms of efficiency, we introduce a second construction that equalizes the *Average Set Size* across groups.

Theorem 5 (Fair CP with Equalized Coverage, Equalized Average Set Size, and Equalized Set Size - Sensitive Feature Known at Test Time) *Assume that the group assignment function $g : \mathcal{X} \rightarrow \mathcal{G}$ is available at test time. For all $a \in \mathcal{G}$, let $\bar{Q}_{1-\alpha}^{(a)}$ denote the group-conditional quantile constructed using the symmetrized score $\bar{s}$ defined in Eq. (11) at confidence level $1 - \alpha \in (0, 1)$. Let $\mathcal{C}_{k_a}^{\text{EQ}}$ denote the conformal predictor obtained by applying a group-specific scaling factor $k_a \geq 1$ to the conformal threshold, i.e.,*

$$\mathcal{C}_{k_a}^{\text{EQ}}(\mathbf{x}) = \left\{ y \in \mathcal{Y} : \bar{s}(f, \mathbf{x}, y) \leq k_a \bar{Q}_{1-\alpha}^{(a)} \right\}. \quad (14)$$

Assume exchangeability. Then $\mathcal{C}_{k_a}^{\text{EQ}}$ satisfies Equalized Coverage, Equalized Average Set Size, and Equalized Set Size.

Proof The proof of Theorem 5 is provided in Appendix A.2. ■

We now turn to the setting in which the group assignment function $g : \mathcal{X} \rightarrow \mathcal{G}$, i.e., the sensitive feature, is available at calibration time but not at test time. In this case, we define a conservative version of the predictor in Eq. (13) by replacing the group-specific threshold with the largest threshold across groups, thereby obtaining a version of Theorem 4 that applies when the sensitive feature is unknown at test time.

Theorem 6 (Fair CP with Equalized Coverage and Equalized Set Size - Sensitive Feature Unknown at Test Time) *Let $\mathcal{C}_{\max}^{\text{EQ}}$ be the conformal predictor defined as*

$$\mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x}) = \left\{ y \in \mathcal{Y} : \bar{s}(f, \mathbf{x}, y) \leq \bar{Q}_{1-\alpha}^{\max} \right\}, \quad \bar{Q}_{1-\alpha}^{\max} = \max_{a \in \mathcal{G}} \bar{Q}_{1-\alpha}^{(a)}. \quad (15)$$

Assume that the data are exchangeable, and let $\bar{s}$ be the symmetrized score defined in Eq. (11). Then $\mathcal{C}_{\max}^{\text{EQ}}$ satisfies both Equalized Coverage and Equalized Set Size.

Proof The proof of Theorem 6 is provided in Appendix A.3. ■

The use of the maximal group-conditional threshold in Theorem 6 yields a CP that is conservative with respect to the predictor in Theorem 4. As a consequence, the marginal coverage of $\mathcal{C}_{\max}^{\text{EQ}}$ is no smaller than that of $\mathcal{C}^{\text{EQ}}$, as stated in the following corollary.

Corollary 7 (Marginal Coverage Increase when the Sensitive Feature is Unknown) *Let $\mathcal{C}^{\text{EQ}}$ and $\mathcal{C}_{\max}^{\text{EQ}}$ be defined as in Theorems 4 and 6, respectively. Under exchangeability,*

$$\mathbb{P} \{y_{n+1} \in \mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x}_{n+1})\} \geq \mathbb{P} \{y_{n+1} \in \mathcal{C}^{\text{EQ}}(\mathbf{x}_{n+1})\}. \quad (16)$$

Proof The proof of Corollary 7 is provided in Appendix A.4. ■

In summary, we have designed a CP framework that combines Mondrian calibration with counterfactual symmetrization, thereby achieving simultaneous group-level validity, namely *Equalized Coverage* and *Equalized Average Set Size*, and individual-level invariance, namely *Equalized Set Size*, as established in Theorems 4 and 5 when the sensitive feature is available at test time. When it is not, we instead consider a group-agnostic construction based on the maximal group-conditional quantile, which preserves *Equalized Coverage* and *Equalized Set Size* but does not allow for *Equalized Average Set Size*, as established in Theorem 6. Indeed, since the predictor relies on a single, group-agnostic threshold, it cannot adapt to group-specific score distributions. As a consequence, the expected size of the prediction sets may differ across groups. For this reason, we now report an impossibility result: when the sensitive attribute is not available at test time, it is not possible, in general, to simultaneously achieve *Equalized Coverage* and *Equalized Average Set Size*.

Theorem 8 (Impossibility of Equalized Average Set Size when the Sensitive Feature is Unknown at Test Time) *Let $\mathcal{C} : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$ be a conformal predictor of the form*

$$\mathcal{C}(\mathbf{x}) = \{y \in \mathcal{Y} : s(f, \mathbf{x}, y) \leq t\}, \quad (17)$$

*for some nonconformity score function $s : \mathcal{F} \times \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ and threshold $t \in \mathbb{R}$ independent of $g(\mathbf{x})$. Assume that the conditional distributions of the scores $s(f, \mathbf{x}, y)$ given $g(\mathbf{x}) = a$ and $g(\mathbf{x}) = \bar{a}$ are not identical. Then, in general, it is not possible to simultaneously achieve *Equalized Coverage* and *Equalized Average Set Size*.*

Proof The proof of Theorem 8 is provided in Appendix A.5. ■

5. Adversarial Setting

In this section, we formalize the adversarial setting considered in this work. Specifically, we study evasion attacks that do not seek to degrade predictive accuracy, but rather to compromise the fairness properties of CP. We begin by defining the threat model and characterizing the adversary’s capabilities and objectives (Section 5.1). We then introduce a novel attack that amplifies both individual- and group-level disparities under bounded perturbations (Section 5.2). Finally, we present a defense strategy based on provably efficient and robust CP, with the goal of preserving coverage and fairness guarantees in the presence of adversarial perturbations (Section 5.3).

5.1. Threat Model

We consider a white-box adversary with full knowledge of the predictive model f , including its architecture and parameters.

At test time, given an input $\mathbf{x} \in \mathcal{X}$ with label $y \in \mathcal{Y}$, the adversary may construct a perturbed version $\tilde{\mathbf{x}} \in \mathcal{P}(\mathbf{x})$.

Consistently with the fairness setting considered in this work, we assume that the sensitive attribute associated with $\mathbf{x}$ is also available to the adversary, i.e., $g(\tilde{\mathbf{x}}) = g(\mathbf{x})$. This allows the attacker to explicitly target disparities across groups or across counterfactual realizations.

The adversary operates within a class of known attack procedures, selecting an attack $\mathbf{A}_\theta \in \mathcal{A}$, where $\mathcal{A}$ may include multiple parameterizations, i.e., values of $\mathbf{A}$, of the same attack⁸ (Mehrabani et al., 2021b; Gittens et al., 2022; Carlevaro et al., 2025). This assumption reflects practical constraints, as it restricts the adversary to implementable and well-studied attacks from the literature, rather than relying on unrealizable or computationally intractable threat models.

Overall, the adversary seeks to construct perturbations that maximize violations of the fairness properties of CP under the prescribed constraints.

5.2. Attack

Under the considered threat model, the adversary seeks to construct bounded perturbations that maximize violations of the fairness properties satisfied by the predictor.

We first consider attacks against counterfactual invariance. Given a model f , an input $\mathbf{x} \in \mathcal{X}$ with label $y \in \mathcal{Y}$, and its counterfactual counterpart $\bar{\mathbf{x}} = c(\mathbf{x})$ the adversary solves

$$A_1 : \max_{\|\delta\|_p \leq \epsilon} |\lambda \ell(f(\mathbf{x} + \delta), y) - (1 - \lambda) \ell(f(\bar{\mathbf{x}} + \delta), y)|, \quad (18)$$

where $\ell : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a loss function measuring the quality of the model predictions and $\lambda \in (0, 1)$ is a weighting parameter that controls the relative importance of the original and counterfactual inputs in the attack objective. This objective searches for perturbations that maximize the discrepancy between the model behavior on $\mathbf{x}$ and on $\bar{\mathbf{x}}$ under the same perturbation, thus directly targeting the symmetry induced by the counterfactual operator. The adversary may also target disparities at the group level by conditioning the attack objective on group membership. Specifically, for a given group $a \in \mathcal{G}$, the adversary restricts attention to inputs $\mathbf{x}$ such that $g(\mathbf{x}) = a$ and solves

$$A_2 : \max_{\|\delta\|_p \leq \epsilon} |\lambda \ell(f(\mathbf{x} + \delta), y) - (1 - \lambda) \ell(f(\bar{\mathbf{x}} + \delta), y)| \quad \text{s.t. } g(\mathbf{x}) = a, \quad (19)$$

while leaving all other inputs unchanged.

If f and ℓ are differentiable, then the well-known optimization algorithms from the adversarial robustness literature originally developed for accuracy-based attacks can also be applied here, such as the Fast Gradient Sign Method (FGSM) (Goodfellow et al., 2014), Projected Gradient Descent (PGD) (Madry et al., 2017; Croce and Hein, 2020), Auto-PGD (APGD) (Croce and Hein, 2020), and AutoAttack (Croce and Hein, 2020). These attacks

8. That is, the same attack procedure may appear in $\mathcal{A}$ under different parameter configurations.

can be naively adapted by simply replacing the attack objective described in Section 3.4 with the objectives A_1 and A_2 in Eqns. (18) and (19), respectively. Note that both A_1 and A_2 may simultaneously affect the model’s accuracy, as measured by standard classification metrics, and its fairness, as measured by differences in coverage and prediction set sizes across groups and counterfactuals.

5.3. Defense

To mitigate the adversarial attacks introduced in Section 5.2, we adopt the provably efficient and robust conformal prediction (PERCP) framework of (Carlevaro et al., 2025, Theorem 1), which provides finite-sample coverage guarantees under worst-case perturbations.

All conformal predictors considered in this work, namely $\mathcal{C}^{\text{EQ}}$ (Theorem 4), $\mathcal{C}_{k_a}^{\text{EQ}}$ (Theorem 5), and $\mathcal{C}_{\max}^{\text{EQ}}$ (Theorem 6), rely on the same nonconformity score, namely the symmetrized score $\bar{s}(f, \mathbf{x}, y)$ defined in Eq. (11), and differ only in the choice of conformal threshold. We first note that $\bar{s}$ satisfies (Carlevaro et al., 2025, Property 1), namely that evasion attacks can only increase the score. Indeed, $\bar{s}(f, \mathbf{x}, y)$ is obtained by symmetrizing a base score $s(f, \mathbf{x}, y)$ through a symmetric aggregation operator; therefore, whenever $s(f, \mathbf{x}, y)$ satisfies (Carlevaro et al., 2025, Property 1), the same property is inherited by $\bar{s}(f, \mathbf{x}, y)$. As a consequence, PERCP can be directly applied to the symmetrized score.

Under the threat model described in Section 5.1, for each input $\mathbf{x}_{n+1}$ with label y_{n+1} , the adversary generates a perturbed sample $\tilde{\mathbf{x}}_{n+1} \in \mathcal{P}(\mathbf{x}_{n+1})$ by approximately solving either A_1 or A_2 in Eqns. (18)-(19). In practice, this induces a family $\mathcal{A}$ of attack implementations associated with A_1 and A_2 . For example, an element $\mathbf{A}_\theta \in \mathcal{A}$ may correspond to a PGD-based approximation of A_1 with hyperparameter configuration θ , such as the number of gradient steps and a specific value of λ . Following PERCP, we define the worst-case attacked sample as

$$\mathbf{x}_{\mathcal{A}}(\mathbf{x}_{n+1}) = \arg \min_{\mathbf{A}_\theta \in \mathcal{A}} \mathbb{F}(\mathbf{A}_\theta(\mathbf{x}_{n+1}, y_{n+1}, f))_y, \quad (20)$$

and build the attacked calibration scores according to the same group-wise structure introduced in Section 3.3:

$$\bar{s}_{\mathcal{A}}^{(a)} = [\bar{s}(f, \mathbf{x}_{\mathcal{A}}(\mathbf{x}), y) : (\mathbf{x}, y) \in \mathcal{D}_{\mathcal{C}}, g(\mathbf{x}) = a], \quad \forall a \in \mathcal{G}. \quad (21)$$

The conformal thresholds are then replaced by their adversarial counterparts. Specifically, for $\mathcal{C}^{\text{EQ}}$ we use $Q_{1-\alpha}(\bar{s}_{\mathcal{A}}^{(a)})$, for $\mathcal{C}_{k_a}^{\text{EQ}}$ we use $k_a Q_{1-\alpha}(\bar{s}_{\mathcal{A}}^{(a)})$, and for $\mathcal{C}_{\max}^{\text{EQ}}$ we use $\max_{a \in \mathcal{G}} Q_{1-\alpha}(\bar{s}_{\mathcal{A}}^{(a)})$. The resulting predictors are obtained by replacing the original thresholds with these adversarially calibrated ones. Formally, for any predictor introduced in Section 3.3, we define its adversarially robust counterpart $\tilde{\mathcal{C}}$ as

$$\tilde{\mathcal{C}}(\tilde{\mathbf{x}}) = \left\{ y \in \mathcal{Y} : \bar{s}(f, \tilde{\mathbf{x}}, y) \leq \tilde{Q}_{1-\alpha}(g(\mathbf{x})) \right\}, \quad (22)$$

where $\tilde{Q}_{1-\alpha}(g(\mathbf{x}))$ denotes the appropriate adversarially calibrated threshold, depending on whether we consider $\mathcal{C}^{\text{EQ}}$, $\mathcal{C}_{k_a}^{\text{EQ}}$, or $\mathcal{C}_{\max}^{\text{EQ}}$.

Theorem 9 (Fair CP with Equalized Coverage in the Adversarial Setting) *Under the threat model of Section 5.1, for any attack $\mathbf{A} \in \mathcal{A}$, the conformal predictors defined*

Table 1: Datasets used in this paper.

Dataset	Citation	# Instances	# Features	Sensitive attribute	ACC	TE
Adult	(Becker and Kohavi, 1996)	48842	14	Sex (Male/Female)	83.80	2.04
Arrhythmia	(Guvenir et al., 1997)	452	279	Sex (Male/Female)	74.72	2.20
Bank Marketing	(Moro et al., 2014)	45211	16	Marital status (Married/Single)	89.72	4.20
COMPAS	(Angwin et al., 2022)	6172	10	Race (White/Non-white)	68.53	5.06
German Credit	(Hofmann, 1994)	1000	20	Foreign (Non-foreigner/Foreigner)	69.90	10.95
Student Performance	(Cortez, 2008)	649	31	Sex (Male/Female)	85.30	1.00

in Eq. (22) satisfy Equalized Coverage at level $1 - \alpha$, that is,

$$\mathbb{P} \left\{ y_{n+1} \in \tilde{\mathcal{C}}(\tilde{\mathbf{x}}_{n+1}) \mid g(\mathbf{x}_{n+1}) = a \right\} \geq 1 - \alpha, \quad \forall a \in \mathcal{G}. \quad (23)$$

Proof The proof of Theorem 9 is provided in Appendix A.6. ■

Because the defense acts exclusively at the level of the calibration distribution, without modifying either the symmetrized score $\bar{s}$ or the structure of the conformal predictors, the fairness properties established in Section 3.3 are preserved under adversarial perturbations. In particular, Equalized Set Size continues to hold, since it depends only on the counterfactual invariance of $\bar{s}$. Likewise, *Equalized Average Set Size*, when enforced through the scaling factors $(k_a)_{a \in \mathcal{G}}$, is preserved because these factors remain unchanged or can be recalculated.

6. Experimental Results

In this section, we evaluate the methodologies proposed in Sections 4 and 5 on a diverse set of fairness-related benchmark datasets.

The datasets considered in this study, summarized in Table 1, include both small- and large-scale problems and span several domains, including credit scoring, healthcare, education, and social prediction. Each dataset is associated with a sensitive attribute, which is used to assess potential disparities in model predictions across demographic groups.

Table 1 also reports natural performance in terms of standard classification accuracy (ACC) and fairness-related metrics, namely the Total Effect (TE), which quantifies overall group disparity⁹.

All models are implemented as standard multilayer perceptrons and are trained and evaluated using our publicly available repository⁴, ensuring full reproducibility of the experimental setup. For all experiments reported in the tables, we set the confidence level to $1 - \alpha = 0.90$ and the attack hyperparameters to $\lambda = 0.6$ and $\varepsilon = 0.25$. We first consider the natural setting, reported in Table 2, and then the adversarial setting. In the adversarial setting, we consider two attack scenarios, as described in Section 5.2: A_1 attacks, applied

9. The *Total Effect* (TE) measures individual-level counterfactual disparity by quantifying the proportion of instances for which the prediction changes under a counterfactual intervention on the sensitive attribute: $\text{TE} = \mathbb{E} [\mathbb{1} \{ \hat{y}(\mathbf{x}) \neq \hat{y}(\tilde{\mathbf{x}}) \}]$.

Table 2: **Fairness evaluation of CP in the natural setting.** Group-wise coverage and average set size for the sensitive groups a and $\bar{a}$, evaluated on factual $\mathbf{x}$ and counterfactual $\bar{\mathbf{x}}$ samples under different CP strategies (mean $\pm$ std over runs).

Dataset		$\mathcal{C}$		$\mathcal{C}^{\text{EQ}}$		$\mathcal{C}_{k_a}^{\text{EQ}}$		$\mathcal{C}_{\text{max}}^{\text{EQ}}$	
		Coverage	Avg Set Size	Coverage	Avg Set Size	Coverage	Avg Set Size	Coverage	Avg Set Size
		$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$	$a/\bar{a}$
Adult	$\mathbf{x}$	0.89/0.90	0.96/1.30	0.89/0.91	0.97/1.26	0.96/0.91	1.27/1.27	0.95/0.91	1.10/1.26
		$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
	$\bar{\mathbf{x}}$	0.89/0.88	1.22/0.95	0.91/0.89	1.26/0.97	0.91/0.96	1.27/1.27	0.91/0.95	1.26/1.10
		$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
Arrhythmia	$\mathbf{x}$	0.85/0.92	1.21/1.75	0.82/0.92	1.18/1.75	0.97/0.92	1.77/1.75	0.95/0.92	1.59/1.75
		$\pm 0.11/0.07$	$\pm 0.13/0.12$	$\pm 0.12/0.07$	$\pm 0.12/0.12$	$\pm 0.05/0.07$	$\pm 0.13/0.12$	$\pm 0.07/0.07$	$\pm 0.15/0.12$
	$\bar{\mathbf{x}}$	0.92/0.82	1.75/1.15	0.92/0.82	1.75/1.18	0.92/0.97	1.75/1.77	0.92/0.95	1.75/1.59
		$\pm 0.07/0.12$	$\pm 0.12/0.11$	$\pm 0.07/0.12$	$\pm 0.12/0.12$	$\pm 0.07/0.05$	$\pm 0.12/0.13$	$\pm 0.07/0.07$	$\pm 0.12/0.15$
Bank Marketing	$\mathbf{x}$	0.91/0.90	1.05/0.98	0.90/0.90	1.03/0.99	0.90/0.93	1.03/1.03	0.90/0.92	1.03/1.03
		$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$
	$\bar{\mathbf{x}}$	0.89/0.91	0.98/1.05	0.90/0.90	0.99/1.03	0.93/0.90	1.03/1.03	0.92/0.90	1.03/1.03
		$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.00/0.01$
COMPAS	$\mathbf{x}$	0.91/0.91	1.59/1.59	0.91/0.90	1.60/1.57	0.91/0.91	1.60/1.60	0.94/0.90	1.71/1.57
		$\pm 0.02/0.03$	$\pm 0.03/0.04$	$\pm 0.02/0.03$	$\pm 0.03/0.04$	$\pm 0.02/0.03$	$\pm 0.03/0.04$	$\pm 0.02/0.03$	$\pm 0.03/0.04$
	$\bar{\mathbf{x}}$	0.87/0.93	1.49/1.69	0.90/0.91	1.57/1.60	0.91/0.91	1.60/1.60	0.90/0.94	1.57/1.71
		$\pm 0.03/0.02$	$\pm 0.04/0.03$	$\pm 0.03/0.02$	$\pm 0.04/0.03$	$\pm 0.03/0.02$	$\pm 0.04/0.03$	$\pm 0.03/0.02$	$\pm 0.04/0.03$
German Credit	$\mathbf{x}$	1.00/0.91	1.78/1.54	1.00/0.92	1.67/1.55	1.00/0.94	1.67/1.66	1.00/0.96	1.67/1.79
		$\pm 0.00/0.04$	$\pm 0.27/0.07$	$\pm 0.00/0.04$	$\pm 0.31/0.07$	$\pm 0.00/0.03$	$\pm 0.31/0.07$	$\pm 0.00/0.03$	$\pm 0.31/0.06$
	$\bar{\mathbf{x}}$	0.90/1.00	1.48/1.89	0.92/1.00	1.55/1.67	0.94/1.00	1.66/1.67	0.96/1.00	1.79/1.67
		$\pm 0.04/0.00$	$\pm 0.07/0.21$	$\pm 0.04/0.00$	$\pm 0.07/0.31$	$\pm 0.03/0.00$	$\pm 0.07/0.31$	$\pm 0.03/0.00$	$\pm 0.06/0.31$
Student Performance	$\mathbf{x}$	0.95/0.89	1.27/1.29	0.95/0.89	1.27/1.27	0.95/0.89	1.27/1.27	0.95/0.89	1.27/1.30
		$\pm 0.04/0.07$	$\pm 0.09/0.11$	$\pm 0.04/0.07$	$\pm 0.09/0.10$	$\pm 0.04/0.07$	$\pm 0.09/0.10$	$\pm 0.04/0.07$	$\pm 0.09/0.11$
	$\bar{\mathbf{x}}$	0.89/0.95	1.26/1.27	0.89/0.95	1.27/1.27	0.89/0.95	1.27/1.27	0.89/0.95	1.30/1.27
		$\pm 0.07/0.04$	$\pm 0.10/0.09$	$\pm 0.07/0.04$	$\pm 0.10/0.09$	$\pm 0.07/0.04$	$\pm 0.10/0.09$	$\pm 0.07/0.04$	$\pm 0.11/0.09$

across the entire population and reported in Table 3, and A_2 group-targeted attacks, specifically designed to affect sensitive subgroups and reported in Table 4.

Finally, due to space constraints and to provide a more detailed empirical analysis, we defer the extensive evaluation of the COMPAS dataset to Appendix B. As the main case study widely adopted in fairness evaluation benchmarks, we analyze it under different perturbation regimes (with ε ranging from 0 to 0.5) and across sensitive groups. The corresponding graphical analyses, illustrating the trends of coverage and average prediction set size under both group-level (A_1) and group-targeted (A_2) attacks, fully support the insights discussed in this section and show how our defense consistently restores target guarantees as the attack strength increases.

Table 2 reports group-conditional coverage and average prediction set size for the standard conformal predictor $\mathcal{C}$ and its fairness-aware variants. Overall, $\mathcal{C}$ already achieves competitive marginal performance but exhibits non-negligible disparities between the protected groups a and $\bar{a}$, particularly in terms of coverage and set size on more challenging datasets. The experimental results are consistent with the theoretical findings. In particular, the rows associated with $\mathbf{x}$ (factual samples) and $\bar{\mathbf{x}}$ (counterfactual samples) confirm that all the proposed methods successfully enforce *Equalized Coverage* and *Equalized Set Size* across the

protected groups. Moreover, the *Equalized Average Set Size* property is empirically verified for the proposed scaling factor k_a , showing that the average prediction set size can be effectively aligned between groups while preserving the desired coverage guarantees. In practice, k_a is estimated on the calibration set through a simple one-dimensional search procedure, implemented as a binary search over the scaling parameter applied to the conformal threshold. The target value is chosen as the average set size of the conformal predictor associated with the group having the largest empirical average prediction set size, and the selected k_a is the one producing the closest average set size for the other group. Differences across methods are generally small on easier datasets, while they become more evident in settings with higher complexity or stronger imbalance (e.g., Arrhythmia and German Credit), highlighting the effectiveness of the proposed adjustments in enforcing more equitable conformal behavior. Table 3 reports the impact of group-level attacks on the group-wise performance of CPs. Across all datasets, attacks consistently affect both coverage and average set size, with heterogeneous effects across sensitive groups. Standard CP already exhibits mild disparities between groups in the clean setting, which are further amplified under adversarial perturbations. While fairness-aware variants partially mitigate these gaps in benign conditions, their robustness under attack varies significantly across datasets. In particular, stronger equalization constraints tend to stabilize coverage disparities, but often at the cost of larger or less balanced prediction sets. Crucially, linking back to the calibration strategy described above for enforcing *Equalized Average Set Size* via the scaling factor k_a , we observe that under attack this balance is disrupted: the target alignment of average set sizes across groups is no longer preserved when the adversary shifts the underlying distribution. Our defense explicitly compensates for this effect, restoring the calibration target defined by the clean setting (i.e., the group with the largest empirical average set size) and re-aligning the scaling procedure accordingly. As a result, it effectively recovers the coverage levels observed in the natural scenario and re-establishes group consistency in terms of fairness metrics, at the cost of a moderate increase in prediction set size under adversarial conditions.

Table 4 reports the impact of group-targeted attacks on the group-wise performance of CPs. In contrast to the group-level attacks in Table 3, these perturbations induce strongly asymmetric effects, consistently degrading coverage for the targeted group while leaving the other group largely unaffected. This asymmetry leads to a marked amplification of fairness violations, with standard CP showing pronounced group-wise discrepancies under attack. Fairness-aware variants provide partial mitigation, but remain sensitive to targeted degradation, particularly when enforcing strict alignment constraints. As in the previous setting, stronger equalization strategies tend to reduce coverage gaps, although often at the expense of larger and more imbalanced prediction sets. In this case, the adversarial effect can be interpreted as a maximization over groups, where the attacker selectively focuses on the most vulnerable group to maximize the disparity. Our defense counters this worst-case behavior by stabilizing coverage for the most affected group, thereby significantly reducing the induced asymmetry and restoring more balanced performance across groups, again with a moderate increase in prediction set size.

Table 3: **Group-wise robustness under group attacks (A_1)**. Group-wise coverage and average set size for the sensitive groups a and $\bar{a}$, evaluated on attacked factual $\tilde{x}$ and attacked counterfactual $\tilde{\bar{x}}$ samples under different CP strategies (mean $\pm$ std over runs). Purple rows highlight the CP setting without defense.

Dataset		$\mathcal{C} - \tilde{\mathcal{C}}$		$\mathcal{C}^{\text{EQ}} - \tilde{\mathcal{C}}^{\text{EQ}}$		$\mathcal{C}_{k_a}^{\text{EQ}} - \tilde{\mathcal{C}}_{k_a}^{\text{EQ}}$		$\mathcal{C}_{\max}^{\text{EQ}} - \tilde{\mathcal{C}}_{\max}^{\text{EQ}}$	
		Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$
Adult	$\tilde{x}$	0.88/0.78	0.96/1.18	0.87/0.84	0.95/1.24	0.93/0.84	1.24/1.24	0.93/0.84	1.10/1.24
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
	$\tilde{x}$	0.84/0.84	1.21/0.94	0.84/0.87	1.24/0.95	0.84/0.93	1.24/1.24	0.84/0.93	1.24/1.10
	$\tilde{\bar{x}}$	$\pm 0.01/0.02$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.02$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
	$\tilde{x}$	0.89/0.90	0.98/1.79	0.89/0.91	1.00/1.48	0.94/0.91	1.48/1.48	0.93/0.91	1.30/1.48
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.01$	$\pm 0.01/0.01$	$\pm 0.02/0.01$
Arrhythmia	$\tilde{x}$	0.96/0.89	1.85/0.99	0.91/0.89	1.48/1.00	0.91/0.94	1.48/1.48	0.91/0.93	1.48/1.30
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.02$	$\pm 0.01/0.01$	$\pm 0.01/0.02$
	$\tilde{x}$	0.67/0.69	1.03/1.52	0.67/0.69	1.03/1.52	0.74/0.69	1.54/1.52	0.67/0.69	1.31/1.52
	$\tilde{\bar{x}}$	$\pm 0.15/0.13$	$\pm 0.05/0.14$	$\pm 0.15/0.13$	$\pm 0.05/0.14$	$\pm 0.14/0.13$	$\pm 0.16/0.14$	$\pm 0.15/0.13$	$\pm 0.14/0.14$
	$\tilde{x}$	0.69/0.67	1.52/1.00	0.69/0.67	1.52/1.03	0.69/0.74	1.52/1.54	0.69/0.67	1.52/1.31
	$\tilde{\bar{x}}$	$\pm 0.13/0.15$	$\pm 0.14/0.00$	$\pm 0.13/0.15$	$\pm 0.14/0.05$	$\pm 0.13/0.14$	$\pm 0.14/0.16$	$\pm 0.13/0.15$	$\pm 0.14/0.14$
Bank Marketing	$\tilde{x}$	0.95/0.94	1.82/1.94	0.95/0.94	1.82/1.94	1.00/1.00	2.00/2.00	0.97/0.94	1.87/1.94
	$\tilde{\bar{x}}$	$\pm 0.07/0.06$	$\pm 0.12/0.06$	$\pm 0.07/0.06$	$\pm 0.12/0.06$	$\pm 0.00/0.00$	$\pm 0.00/0.00$	$\pm 0.05/0.06$	$\pm 0.10/0.06$
	$\tilde{x}$	0.94/0.95	1.94/1.82	0.94/0.95	1.94/1.82	1.00/1.00	2.00/2.00	0.94/0.97	1.94/1.87
	$\tilde{\bar{x}}$	$\pm 0.06/0.07$	$\pm 0.06/0.12$	$\pm 0.06/0.07$	$\pm 0.06/0.12$	$\pm 0.00/0.00$	$\pm 0.00/0.00$	$\pm 0.06/0.05$	$\pm 0.06/0.10$
	$\tilde{x}$	0.88/0.90	1.06/1.00	0.88/0.87	1.04/0.98	0.88/0.91	1.04/1.04	0.88/0.91	1.04/1.04
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
COMPAS	$\tilde{x}$	0.90/0.90	0.99/1.08	0.87/0.88	0.98/1.04	0.91/0.88	1.04/1.04	0.91/0.88	1.04/1.04
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
	$\tilde{x}$	0.91/0.90	1.56/0.99	0.91/0.90	1.07/1.03	0.91/0.93	1.07/1.07	0.91/0.94	1.07/1.07
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.02/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.00$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
	$\tilde{x}$	0.89/0.94	0.98/1.49	0.90/0.91	1.03/1.07	0.93/0.91	1.07/1.07	0.94/0.91	1.07/1.07
	$\tilde{\bar{x}}$	$\pm 0.01/0.01$	$\pm 0.00/0.02$	$\pm 0.01/0.01$	$\pm 0.00/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$	$\pm 0.01/0.01$
German Credit	$\tilde{x}$	0.70/0.75	1.44/1.44	0.82/0.78	1.54/1.45	0.83/0.79	1.54/1.54	0.83/0.78	1.61/1.45
	$\tilde{\bar{x}}$	$\pm 0.03/0.04$	$\pm 0.03/0.04$	$\pm 0.02/0.04$	$\pm 0.03/0.04$	$\pm 0.02/0.04$	$\pm 0.03/0.04$	$\pm 0.02/0.04$	$\pm 0.03/0.04$
	$\tilde{x}$	0.79/0.86	1.44/1.64	0.78/0.82	1.45/1.54	0.79/0.83	1.54/1.54	0.78/0.83	1.45/1.61
	$\tilde{\bar{x}}$	$\pm 0.04/0.02$	$\pm 0.04/0.03$	$\pm 0.04/0.02$	$\pm 0.04/0.03$	$\pm 0.04/0.02$	$\pm 0.04/0.03$	$\pm 0.04/0.02$	$\pm 0.04/0.03$
	$\tilde{x}$	0.89/0.91	1.84/1.88	0.89/0.90	1.82/1.87	0.92/0.90	1.87/1.87	0.92/0.90	1.88/1.87
	$\tilde{\bar{x}}$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$	$\pm 0.02/0.03$
Student Performance	$\tilde{x}$	0.93/0.93	1.90/1.88	0.90/0.89	1.87/1.82	0.90/0.92	1.87/1.87	0.90/0.92	1.87/1.88
	$\tilde{\bar{x}}$	$\pm 0.02/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$	$\pm 0.03/0.02$
	$\tilde{x}$	0.78/0.82	1.56/1.47	0.78/0.85	1.44/1.48	0.78/0.85	1.56/1.48	0.78/0.87	1.44/1.70
	$\tilde{\bar{x}}$	$\pm 0.27/0.05$	$\pm 0.32/0.07$	$\pm 0.27/0.05$	$\pm 0.32/0.07$	$\pm 0.27/0.05$	$\pm 0.32/0.07$	$\pm 0.27/0.05$	$\pm 0.32/0.06$
	$\tilde{x}$	0.72/0.78	1.31/1.67	0.85/0.78	1.48/1.44	0.85/0.78	1.48/1.56	0.87/0.78	1.70/1.44
	$\tilde{\bar{x}}$	$\pm 0.06/0.27$	$\pm 0.06/0.31$	$\pm 0.05/0.27$	$\pm 0.07/0.32$	$\pm 0.05/0.27$	$\pm 0.07/0.32$	$\pm 0.05/0.27$	$\pm 0.06/0.32$
Student Performance	$\tilde{x}$	0.89/0.86	1.89/1.75	0.89/0.87	1.89/1.80	0.89/0.89	1.89/1.89	0.89/0.94	1.89/1.94
	$\tilde{\bar{x}}$	$\pm 0.21/0.05$	$\pm 0.21/0.06$	$\pm 0.21/0.05$	$\pm 0.21/0.06$	$\pm 0.21/0.04$	$\pm 0.21/0.04$	$\pm 0.21/0.03$	$\pm 0.21/0.03$
	$\tilde{x}$	0.84/0.78	1.71/1.78	0.87/0.89	1.80/1.89	0.89/0.89	1.89/1.89	0.94/0.89	1.94/1.89
	$\tilde{\bar{x}}$	$\pm 0.05/0.27$	$\pm 0.06/0.27$	$\pm 0.05/0.21$	$\pm 0.06/0.21$	$\pm 0.04/0.21$	$\pm 0.04/0.21$	$\pm 0.03/0.21$	$\pm 0.03/0.21$
	$\tilde{x}$	0.66/0.56	1.02/1.01	0.67/0.56	1.03/1.01	0.67/0.59	1.03/1.04	0.67/0.59	1.03/1.04
	$\tilde{\bar{x}}$	$\pm 0.10/0.12$	$\pm 0.03/0.03$	$\pm 0.10/0.12$	$\pm 0.04/0.03$	$\pm 0.10/0.12$	$\pm 0.04/0.05$	$\pm 0.10/0.12$	$\pm 0.04/0.05$
Student Performance	$\tilde{x}$	0.56/0.67	1.01/1.03	0.56/0.67	1.01/1.03	0.59/0.67	1.04/1.03	0.59/0.67	1.04/1.03
	$\tilde{\bar{x}}$	$\pm 0.12/0.10$	$\pm 0.03/0.04$	$\pm 0.12/0.10$	$\pm 0.03/0.04$	$\pm 0.12/0.10$	$\pm 0.05/0.04$	$\pm 0.12/0.10$	$\pm 0.05/0.04$
	$\tilde{x}$	0.95/0.89	1.88/1.77	0.95/0.89	1.88/1.77	0.95/0.91	1.88/1.89	0.95/0.89	1.88/1.81
	$\tilde{\bar{x}}$	$\pm 0.04/0.07$	$\pm 0.07/0.10$	$\pm 0.04/0.07$	$\pm 0.07/0.10$	$\pm 0.04/0.07$	$\pm 0.07/0.07$	$\pm 0.04/0.07$	$\pm 0.07/0.09$
	$\tilde{x}$	0.89/0.95	1.76/1.91	0.89/0.95	1.77/1.88	0.91/0.95	1.89/1.88	0.89/0.95	1.81/1.88
	$\tilde{\bar{x}}$	$\pm 0.07/0.04$	$\pm 0.10/0.06$	$\pm 0.07/0.04$	$\pm 0.10/0.07$	$\pm 0.07/0.04$	$\pm 0.07/0.07$	$\pm 0.07/0.04$	$\pm 0.09/0.07$

Table 4: **Group-wise robustness under targeted attacks (A_2)**. Coverage and average prediction set size are reported for the sensitive groups a and $\bar{a}$, evaluated on attacked factual $\tilde{x}$ and attacked counterfactual $\bar{\tilde{x}}$ samples under group-targeted adversarial attacks ($a = 0$ and $a = 1$). Values are averaged over multiple runs (mean over runs) but std is not reported due to space constraints. Purple rows highlight the CP setting without defense.

		$\mathcal{C} - \bar{\mathcal{C}}$		$\mathcal{C}^{\text{EQ}} - \bar{\mathcal{C}}^{\text{EQ}}$		$\mathcal{C}_{k_a}^{\text{EQ}} - \bar{\mathcal{C}}_{k_a}^{\text{EQ}}$		$\mathcal{C}_{\max}^{\text{EQ}} - \bar{\mathcal{C}}_{\max}^{\text{EQ}}$		
		Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	Coverage $a/\bar{a}$	Avg Set Size $a/\bar{a}$	
Adult	$a = 0$	$\tilde{x}$	0.87 / 0.90	0.96 / 1.30	0.87 / 0.91	0.95 / 1.26	0.93 / 0.91	1.27 / 1.27	0.93 / 0.91	1.10 / 1.26
		$\bar{\tilde{x}}$	0.89 / 0.88	1.12 / 0.95	0.91 / 0.87	1.26 / 0.95	0.91 / 0.93	1.27 / 1.27	0.91 / 0.93	1.10 / 1.26
		$\tilde{x}$	0.89 / 0.90	0.98 / 1.30	0.89 / 0.91	1.01 / 1.26	0.93 / 0.91	1.27 / 1.27	0.93 / 0.91	1.10 / 1.26
		$\bar{\tilde{x}}$	0.89 / 0.89	1.22 / 0.99	0.91 / 0.89	1.26 / 1.01	0.91 / 0.93	1.27 / 1.27	0.91 / 0.93	1.26 / 1.10
	$a = 1$	$\tilde{x}$	0.89 / 0.78	0.96 / 1.18	0.89 / 0.84	0.97 / 1.24	0.96 / 0.84	1.23 / 1.24	0.95 / 0.84	1.10 / 1.24
		$\bar{\tilde{x}}$	0.84 / 0.88	1.21 / 0.95	0.84 / 0.89	1.24 / 0.97	0.84 / 0.96	1.24 / 1.23	0.84 / 0.95	1.24 / 1.10
		$\tilde{x}$	0.89 / 0.90	0.96 / 1.78	0.89 / 0.91	0.97 / 1.48	0.96 / 0.91	1.48 / 1.48	0.96 / 0.91	1.33 / 1.48
		$\bar{\tilde{x}}$	0.96 / 0.88	1.84 / 0.95	0.91 / 0.89	1.48 / 0.97	0.91 / 0.96	1.48 / 1.48	0.91 / 0.96	1.48 / 1.33
Arrhythmia	$a = 0$	$\tilde{x}$	0.67 / 0.92	1.03 / 1.75	0.67 / 0.92	1.03 / 1.75	0.87 / 0.92	1.74 / 1.75	0.67 / 0.92	1.31 / 1.75
		$\bar{\tilde{x}}$	0.92 / 0.67	1.75 / 1.00	0.92 / 0.67	1.75 / 1.03	0.92 / 0.87	1.75 / 1.74	0.92 / 0.67	1.75 / 1.31
		$\tilde{x}$	0.95 / 0.92	1.82 / 1.75	0.95 / 0.92	1.82 / 1.75	1.00 / 1.00	2.00 / 2.00	0.95 / 1.00	1.82 / 1.98
		$\bar{\tilde{x}}$	0.92 / 0.95	1.75 / 1.82	0.92 / 0.95	1.75 / 1.82	1.00 / 1.00	2.00 / 2.00	1.00 / 0.95	1.98 / 1.82
	$a = 1$	$\tilde{x}$	0.85 / 0.69	1.21 / 1.52	0.82 / 0.69	1.18 / 1.52	0.95 / 0.69	1.51 / 1.52	0.95 / 0.69	1.59 / 1.52
		$\bar{\tilde{x}}$	0.69 / 0.82	1.52 / 1.15	0.69 / 0.82	1.52 / 1.18	0.69 / 0.95	1.52 / 1.51	0.69 / 0.95	1.52 / 1.59
		$\tilde{x}$	0.85 / 0.94	1.21 / 1.94	0.82 / 0.94	1.18 / 1.94	1.00 / 1.00	2.00 / 2.00	1.00 / 0.94	1.90 / 1.94
		$\bar{\tilde{x}}$	0.94 / 0.82	1.94 / 1.15	0.94 / 0.82	1.94 / 1.18	1.00 / 1.00	2.00 / 2.00	0.94 / 1.00	1.94 / 1.90
Bank Marketing	$a = 0$	$\tilde{x}$	0.88 / 0.90	1.05 / 0.98	0.88 / 0.90	1.04 / 0.99	0.88 / 0.93	1.04 / 1.04	0.88 / 0.92	1.04 / 1.03
		$\bar{\tilde{x}}$	0.89 / 0.90	0.98 / 1.08	0.90 / 0.88	0.99 / 1.04	0.93 / 0.88	1.04 / 1.04	0.92 / 0.88	1.03 / 1.04
		$\tilde{x}$	0.91 / 0.90	1.58 / 0.98	0.91 / 0.90	1.07 / 0.99	0.91 / 0.94	1.07 / 1.07	0.91 / 0.93	1.07 / 1.03
		$\bar{\tilde{x}}$	0.89 / 0.95	0.98 / 1.51	0.90 / 0.91	0.99 / 1.07	0.94 / 0.91	1.07 / 1.07	0.93 / 0.91	1.03 / 1.07
	$a = 1$	$\tilde{x}$	0.91 / 0.90	1.05 / 1.00	0.90 / 0.87	1.03 / 0.98	0.90 / 0.90	1.03 / 1.03	0.90 / 0.91	1.03 / 1.04
		$\bar{\tilde{x}}$	0.90 / 0.91	0.99 / 1.05	0.87 / 0.90	0.98 / 1.03	0.90 / 0.90	1.03 / 1.03	0.91 / 0.90	1.04 / 1.03
		$\tilde{x}$	0.91 / 0.90	1.05 / 0.99	0.90 / 0.90	1.03 / 1.03	0.90 / 0.90	1.03 / 1.03	0.90 / 0.91	1.03 / 1.04
		$\bar{\tilde{x}}$	0.89 / 0.91	0.98 / 1.05	0.90 / 0.90	1.03 / 1.03	0.90 / 0.90	1.03 / 1.03	0.91 / 0.90	1.04 / 1.03
COMPAS	$a = 0$	$\tilde{x}$	0.70 / 0.91	1.44 / 1.59	0.82 / 0.90	1.54 / 1.57	0.83 / 0.90	1.57 / 1.57	0.83 / 0.90	1.61 / 1.57
		$\bar{\tilde{x}}$	0.87 / 0.86	1.49 / 1.64	0.90 / 0.82	1.57 / 1.54	0.90 / 0.83	1.57 / 1.57	0.90 / 0.83	1.57 / 1.61
		$\tilde{x}$	0.89 / 0.91	1.84 / 1.59	0.89 / 0.90	1.82 / 1.57	0.89 / 0.96	1.82 / 1.82	0.89 / 0.98	1.82 / 1.92
		$\bar{\tilde{x}}$	0.87 / 0.93	1.49 / 1.88	0.90 / 0.89	1.57 / 1.82	0.96 / 0.89	1.82 / 1.82	0.98 / 0.89	1.92 / 1.82
	$a = 1$	$\tilde{x}$	0.91 / 0.75	1.59 / 1.44	0.91 / 0.78	1.60 / 1.45	0.91 / 0.80	1.60 / 1.60	0.94 / 0.78	1.71 / 1.45
		$\bar{\tilde{x}}$	0.79 / 0.93	1.44 / 1.69	0.78 / 0.91	1.45 / 1.60	0.80 / 0.91	1.60 / 1.60	0.78 / 0.94	1.45 / 1.71
		$\tilde{x}$	0.91 / 0.91	1.76 / 1.88	0.91 / 0.90	1.60 / 1.87	0.97 / 0.90	1.87 / 1.87	0.99 / 0.90	1.95 / 1.87
		$\bar{\tilde{x}}$	0.93 / 0.91	1.90 / 1.69	0.90 / 0.91	1.87 / 1.60	0.90 / 0.97	1.87 / 1.87	0.90 / 0.99	1.87 / 1.95
German Credit	$a = 0$	$\tilde{x}$	0.78 / 0.91	1.56 / 1.54	0.78 / 0.92	1.44 / 1.55	0.78 / 0.92	1.56 / 1.56	0.78 / 0.96	1.44 / 1.79
		$\bar{\tilde{x}}$	0.90 / 0.78	1.48 / 1.67	0.92 / 0.78	1.55 / 1.44	0.92 / 0.78	1.56 / 1.56	0.96 / 0.78	1.79 / 1.44
		$\tilde{x}$	0.89 / 0.91	1.89 / 1.54	0.89 / 0.92	1.89 / 1.55	0.89 / 0.99	1.89 / 1.89	0.89 / 1.00	1.89 / 2.00
		$\bar{\tilde{x}}$	0.90 / 0.78	1.48 / 1.78	0.92 / 0.89	1.55 / 1.89	0.99 / 0.89	1.89 / 1.89	1.00 / 0.89	2.00 / 1.89
	$a = 1$	$\tilde{x}$	1.00 / 0.82	1.78 / 1.47	1.00 / 0.85	1.67 / 1.48	1.00 / 0.87	1.67 / 1.66	1.00 / 0.87	1.67 / 1.70
		$\bar{\tilde{x}}$	0.72 / 1.00	1.31 / 1.89	0.85 / 1.00	1.48 / 1.67	0.87 / 1.00	1.66 / 1.67	0.87 / 1.00	1.70 / 1.67
		$\tilde{x}$	1.00 / 0.86	1.78 / 1.75	1.00 / 0.87	1.67 / 1.80	1.00 / 0.87	1.78 / 1.81	1.00 / 0.87	1.89 / 1.80
		$\bar{\tilde{x}}$	0.84 / 1.00	1.71 / 1.89	0.87 / 1.00	1.80 / 1.67	0.87 / 1.00	1.81 / 1.78	0.87 / 1.00	1.80 / 1.89
Student Performance	$a = 0$	$\tilde{x}$	0.66 / 0.89	1.02 / 1.29	0.67 / 0.89	1.03 / 1.27	0.79 / 0.89	1.27 / 1.27	0.67 / 0.89	1.03 / 1.30
		$\bar{\tilde{x}}$	0.89 / 0.67	1.26 / 1.03	0.89 / 0.67	1.27 / 1.03	0.89 / 0.79	1.27 / 1.27	0.89 / 0.67	1.30 / 1.03
		$\tilde{x}$	0.95 / 0.89	1.88 / 1.29	0.95 / 0.89	1.88 / 1.27	0.95 / 0.94	1.88 / 1.87	0.95 / 0.96	1.88 / 1.89
		$\bar{\tilde{x}}$	0.89 / 0.95	1.26 / 1.91	0.89 / 0.95	1.27 / 1.88	0.94 / 0.95	1.87 / 1.88	0.96 / 0.95	1.89 / 1.88
	$a = 1$	$\tilde{x}$	0.95 / 0.56	1.27 / 1.01	0.95 / 0.56	1.27 / 1.01	0.95 / 0.76	1.27 / 1.27	0.95 / 0.59	1.27 / 1.04
		$\bar{\tilde{x}}$	0.56 / 0.95	1.01 / 1.27	0.56 / 0.95	1.01 / 1.27	0.76 / 0.95	1.27 / 1.27	0.59 / 0.95	1.04 / 1.27
		$\tilde{x}$	0.95 / 0.89	1.27 / 1.77	0.95 / 0.89	1.27 / 1.77	0.98 / 0.89	1.78 / 1.77	0.98 / 0.89	1.78 / 1.77
		$\bar{\tilde{x}}$	0.89 / 0.95	1.76 / 1.27	0.89 / 0.95	1.77 / 1.27	0.89 / 0.98	1.77 / 1.78	0.89 / 0.98	1.77 / 1.78

7. Conclusions

In this paper, we studied conformal prediction under the joint requirements of uncertainty quantification, fairness, and adversarial robustness. We showed that coverage guarantees alone are not sufficient to ensure fair treatment, and that fairness guarantees obtained in the natural setting may be fragile under adversarial perturbations.

In the natural setting, we proposed conformal predictors that combine counterfactual score symmetrization with group-conditional calibration. When the sensitive attribute is available at test time, the resulting methods can simultaneously enforce individual-level fairness, through *Equalized Set Size*, and group-level fairness, through *Equalized Coverage* and *Equalized Average Set Size*. When the sensitive attribute is unavailable at test time, we showed that a conservative group-agnostic construction based on the largest group-specific quantile still preserves coverage and counterfactual invariance, but may induce over-coverage and larger prediction sets. Our impossibility result formalizes this limitation, showing that exact simultaneous control of *Equalized Coverage* and *Equalized Average Set Size* cannot, in general, be achieved without access to the sensitive attribute at deployment.

We then considered adversarial settings in which perturbations are explicitly designed to amplify fairness violations. The proposed attacks show that both standard and fairness-aware conformal predictors can be vulnerable: global attacks progressively degrade coverage, while targeted attacks induce asymmetric effects across sensitive groups. To address this issue, we introduced an adversarially calibrated defense based on provably efficient and robust conformal prediction. Under the specified threat model, the defense restores coverage guarantees for attacked samples and reduces the disparities induced by fairness-targeted perturbations, at the cost of larger prediction sets.

Overall, our results show that fairness, robustness, and efficiency in CP are tightly coupled. Fairness constraints can promote more equitable behavior, but do not by themselves ensure adversarial robustness; robust calibration can recover validity under attack, but necessarily increases conservativeness. Future work should further characterize this trade-off, develop less conservative adaptive calibration strategies, and extend the framework to broader sensitive-attribute settings, prediction tasks, and adversarial threat models.

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Appendix A. Proofs

A.1. Proof of Theorem 4

Proof We exploit the invariance of the symmetrized score and the group-conditional validity of Mondrian CP. Since Agg is symmetric, the symmetrized score is invariant under the counterfactual operator c (Guldogan et al., 2025, Lemma 4.1), namely,

$$\bar{s}(f, \mathbf{x}, y) = \text{Agg}(s(f, \mathbf{x}, y), s(f, \bar{\mathbf{x}}, y)) = \text{Agg}(s(f, \bar{\mathbf{x}}, y), s(f, \mathbf{x}, y)) = \bar{s}(f, \bar{\mathbf{x}}, y). \quad (24)$$

Therefore, for any $\mathbf{x} \in \mathcal{X}$ and $y \in \mathcal{Y}$,

$$y \in \mathcal{C}^{\text{EQ}}(\mathbf{x}) \Leftrightarrow \bar{s}(f, \mathbf{x}, y) \leq \bar{Q}_{1-\alpha}^{(g(\mathbf{x}))} \Leftrightarrow \bar{s}(f, \bar{\mathbf{x}}, y) \leq \bar{Q}_{1-\alpha}^{(g(\bar{\mathbf{x}}))} \Leftrightarrow y \in \mathcal{C}^{\text{EQ}}(\bar{\mathbf{x}}), \quad (25)$$

which implies $\mathcal{C}^{\text{EQ}}(\mathbf{x}) = \mathcal{C}^{\text{EQ}}(\bar{\mathbf{x}})$ and therefore $|\mathcal{C}^{\text{EQ}}(\mathbf{x})| = |\mathcal{C}^{\text{EQ}}(\bar{\mathbf{x}})|$, i.e., *Equalized Set Size* holds.

By construction, $\mathcal{C}^{\text{EQ}}$ is a Mondrian CP based on $\bar{s}$ with group-conditional quantiles $\bar{Q}_{1-\alpha}^{(a)}$. Under exchangeability, the symmetrized scores are exchangeable (Guldogan et al., 2025, Theorem 4.1); hence, standard Mondrian CP guarantees apply (Vovk et al., 2003), yielding, for all $a \in \mathcal{G}$,

$$\mathbb{P} \{y_{n+1} \in \mathcal{C}^{\text{EQ}}(\mathbf{x}_{n+1}) : g(\mathbf{x}_{n+1}) = a\} \geq 1 - \alpha. \quad (26)$$

This proves *Equalized Coverage*. ■

A.2. Proof of Theorem 5

Proof For every $a \in \mathcal{G}$, $\mathcal{C}^{\text{EQ}}$ is, by construction, a Mondrian CP based on the symmetrized score $\bar{s}$ and the group-conditional quantiles $\bar{Q}_{1-\alpha}^{(a)}$. Under exchangeability, the calibration scores within each group are exchangeable. Since $k_a \geq 1$, we have

$$\bar{Q}_{1-\alpha}^{(a)} \leq k_a \bar{Q}_{1-\alpha}^{(a)}, \quad (27)$$

which implies

$$\mathcal{C}^{\text{EQ}}(\mathbf{x}) \subseteq \mathcal{C}_{k_a}^{\text{EQ}}(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{X}. \quad (28)$$

By monotonicity of probability, it follows that

$$\mathbb{P} \{y_{n+1} \in \mathcal{C}_{k_a}^{\text{EQ}}(\mathbf{x}_{n+1}) : g(\mathbf{x}_{n+1}) = a\} \geq 1 - \alpha, \quad (29)$$

for all $a \in \mathcal{G}$. Hence, *Equalized Coverage* holds.

Moreover, since the aggregation operator Agg is symmetric, the score $\bar{s}$ is invariant under the counterfactual map. Therefore,

$$\mathcal{C}_{k_a}^{\text{EQ}}(\mathbf{x}) = \mathcal{C}_{k_a}^{\text{EQ}}(\bar{\mathbf{x}}), \quad \forall \mathbf{x} \in \mathcal{X}, \quad (30)$$

which proves *Equalized Set Size*.

It remains to establish *Equalized Average Set Size*. For each group $a \in \mathcal{G}$, define

$$E_a(k) := \mathbb{E}[|\mathcal{C}_k^{\text{EQ}}(\mathbf{x})| : g(\mathbf{x}) = a]. \quad (31)$$

We first observe that, for every $a \in \mathcal{G}$, the function $E_a(k)$ is non-decreasing in k . Indeed, for $k_1 \leq k_2$,

$$\mathcal{C}_{k_1}^{\text{EQ}}(\mathbf{x}) \subseteq \mathcal{C}_{k_2}^{\text{EQ}}(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{X}, \quad (32)$$

which implies $E_a(k_1) \leq E_a(k_2)$.

Now let $a, \bar{a} \in \mathcal{G}$. Without loss of generality, assume that

$$E_a(1) \leq E_{\bar{a}}(1). \quad (33)$$

Fix $k_{\bar{a}} = 1$ and define $E = E_{\bar{a}}(1)$. Since $E_a(k)$ is non-decreasing, one can choose $k_a \geq 1$ so that

$$E_a(k_a) = E, \quad (34)$$

or, more generally, so that the average set sizes are matched across groups.¹⁰ Thus,

$$\mathbb{E}[|\mathcal{C}_{k_a}^{\text{EQ}}(\mathbf{x}_{n+1})| : g(\mathbf{x}_{n+1}) = a] = \mathbb{E}[|\mathcal{C}_{k_{\bar{a}}}^{\text{EQ}}(\mathbf{x}_{n+1})| : g(\mathbf{x}_{n+1}) = \bar{a}], \quad (35)$$

which proves *Equalized Average Set Size*. ■

A.3. Proof of Theorem 6

Proof By construction, $\mathcal{C}_{\max}^{\text{EQ}}$ is obtained by thresholding the symmetrized score $\bar{s}$ using the maximal group-conditional quantile

$$\bar{Q}_{1-\alpha}^{\max} = \max_{a \in \mathcal{G}} \bar{Q}_{1-\alpha}^{(a)}. \quad (36)$$

From Theorem 4, the predictor $\mathcal{C}^{\text{EQ}}$ achieves Equalized Coverage at level $1 - \alpha$ when each group uses its own threshold $\bar{Q}_{1-\alpha}^{(a)}$. Since, for every $a \in \mathcal{G}$,

$$\bar{Q}_{1-\alpha}^{\max} \geq \bar{Q}_{1-\alpha}^{(a)}, \quad (37)$$

it follows that, for any $(\mathbf{x}, y)$,

$$\bar{s}(f, \mathbf{x}, y) \leq \bar{Q}_{1-\alpha}^{(a)} \implies \bar{s}(f, \mathbf{x}, y) \leq \bar{Q}_{1-\alpha}^{\max}. \quad (38)$$

Therefore, for all $\mathbf{x} \in \mathcal{X}$,

$$\mathcal{C}^{\text{EQ}}(\mathbf{x}) \subseteq \mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x}). \quad (39)$$

10. If exact equality is not attainable because of discreteness, one may interpret the result in the standard approximate sense or use randomized tie-breaking.

Hence, for every group $a \in \mathcal{G}$, the coverage achieved by $\mathcal{C}_{\max}^{\text{EQ}}$ is at least as large as that achieved by $\mathcal{C}^{\text{EQ}}$, and the conditional coverage guarantee at level $1 - \alpha$ is preserved. This proves *Equalized Coverage*.

The proof of *Equalized Set Size* follows directly from the invariance of the symmetrized score. Since $\mathcal{C}_{\max}^{\text{EQ}}$ is obtained by thresholding $\bar{s}$ with a scalar threshold $\bar{Q}_{1-\alpha}^{\max}$ that does not depend on $\mathbf{x}$, we have

$$\mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x}) = \mathcal{C}_{\max}^{\text{EQ}}(\bar{\mathbf{x}}), \quad \forall \mathbf{x} \in \mathcal{X}. \quad (40)$$

Consequently,

$$|\mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x})| = |\mathcal{C}_{\max}^{\text{EQ}}(\bar{\mathbf{x}})|, \quad \forall \mathbf{x} \in \mathcal{X}, \quad (41)$$

which establishes *Equalized Set Size*. $\blacksquare$

A.4. Proof of Corollary 7

Proof For every $\mathbf{x} \in \mathcal{X}$, we have

$$\mathcal{C}^{\text{EQ}}(\mathbf{x}) \subseteq \mathcal{C}_{\max}^{\text{EQ}}(\mathbf{x}), \quad (42)$$

because $\bar{Q}_{1-\alpha}^{\max} \geq \bar{Q}_{1-\alpha}^{(a)}$ for all $a \in \mathcal{G}$. The claim then follows by monotonicity of probability. $\blacksquare$

A.5. Proof of Theorem 8

Proof We argue by contradiction. Assume that there exists a group-agnostic CP $\mathcal{C}(\mathbf{x})$ with a single threshold t independent of $g(\mathbf{x})$ that simultaneously satisfies: (i) *Equalized Coverage* at level $1 - \alpha$ for all $a \in \mathcal{G}$, and (ii) *Equalized Average Set Size* across groups. *Equalized Coverage* implies that

$$\mathbb{P}\{s(f, \mathbf{x}_{n+1}, y_{n+1}) \leq t : g(\mathbf{x}_{n+1}) = a\} \geq 1 - \alpha, \quad \forall a \in \mathcal{G}. \quad (43)$$

Hence, the threshold t must satisfy

$$t \geq \max_{a \in \mathcal{G}} Q_{1-\alpha}^{(a)}, \quad (44)$$

where $Q_{1-\alpha}^{(a)}$ denotes the $(1 - \alpha)$ -quantile of the score distribution in group a . Let $a^* \in \mathcal{G}$ be a group attaining the maximum, i.e.,

$$Q_{1-\alpha}^{(a^*)} = \max_{a \in \mathcal{G}} Q_{1-\alpha}^{(a)}. \quad (45)$$

Since the conditional score distributions differ across groups, there exists at least one group $\bar{a} \neq a^*$ such that

$$Q_{1-\alpha}^{(\bar{a})} < Q_{1-\alpha}^{(a^*)} \leq t. \quad (46)$$

By definition of quantile,

$$\mathbb{P}\{s(f, \mathbf{x}, y) \leq Q_{1-\alpha}^{(\bar{a})} : g(\mathbf{x}) = \bar{a}\} = 1 - \alpha, \quad (47)$$

and therefore, since $t > Q_{1-\alpha}^{(\bar{a})}$,

$$\mathbb{P}\{s(f, \mathbf{x}, y) \leq t : g(\mathbf{x}) = \bar{a}\} > 1 - \alpha. \quad (48)$$

On the other hand, for the group a^* , we only have

$$\mathbb{P}\{s(f, \mathbf{x}, y) \leq t : g(\mathbf{x}) = a^*\} \geq 1 - \alpha, \quad (49)$$

with equality when $t = Q_{1-\alpha}^{(a^*)}$. Hence, there is a strict imbalance in conditional coverage:

$$\mathbb{P}\{s(f, \mathbf{x}, y) \leq t : g(\mathbf{x}) = \bar{a}\} > \mathbb{P}\{s(f, \mathbf{x}, y) \leq t : g(\mathbf{x}) = a^*\}. \quad (50)$$

Because the conformal set is obtained by thresholding the same score with a common threshold t , this mismatch in group-conditional score distributions induces different expected set sizes across groups. Therefore,

$$\mathbb{E}[|\mathcal{C}(\mathbf{x})| : g(\mathbf{x}) = \bar{a}] \neq \mathbb{E}[|\mathcal{C}(\mathbf{x})| : g(\mathbf{x}) = a^*], \quad (51)$$

which contradicts *Equalized Average Set Size*. This concludes the proof. $\blacksquare$

A.6. Proof of Theorem 9

Proof The claim follows by applying Theorem 1 of Carlevaro et al. (2025) to the symmetrized score $\bar{s}$, which satisfies (Carlevaro et al., 2025, Property 1). Since calibration is performed on worst-case attacked samples $\mathbf{x}_{\mathcal{A}}(\mathbf{x})$, the resulting thresholds control the distribution of adversarial nonconformity scores uniformly over all attacks $\mathbf{A}_\theta \in \mathcal{A}$, thereby yielding valid coverage guarantees. The group-wise construction then ensures that such guarantees hold conditionally on $g(\mathbf{x}) = a$, in accordance with standard Mondrian CP results (Vovk et al., 2003). $\blacksquare$

Appendix B. Detailed Empirical Analysis on the COMPAS Dataset

In this appendix, we provide an extended empirical analysis of the COMPAS dataset, which serves as our primary case study due to its widespread adoption in the fairness literature. Specifically, we evaluate the behavior of group-conditional coverage and average prediction set size as a function of the attack strength ε (ranging from 0 to 0.5), considering both global and group-wise metrics. Stars in the plots denote the specific operating point at $\varepsilon = 0.25$, which corresponds to the values reported in Table 3 and Table 4 of the main manuscript.

B.1. Behavior under Group-Level Attacks (A_1)

Figures 1 and 2 illustrate the performance under group attacks (A_1). As the attack strength ε increases, both coverage and average set size exhibit a monotonic downward trend, reflecting the progressive degradation induced by the adversarial perturbation. This effect is particularly pronounced for the standard CP. In contrast, the fairness-aware variants show improved stability in terms of coverage gaps, though at the cost of consistently larger prediction sets. Across the entire regime, our proposed defense successfully restores coverage close to the nominal level ($1 - \alpha = 0.90$), with only minor residual oscillations.

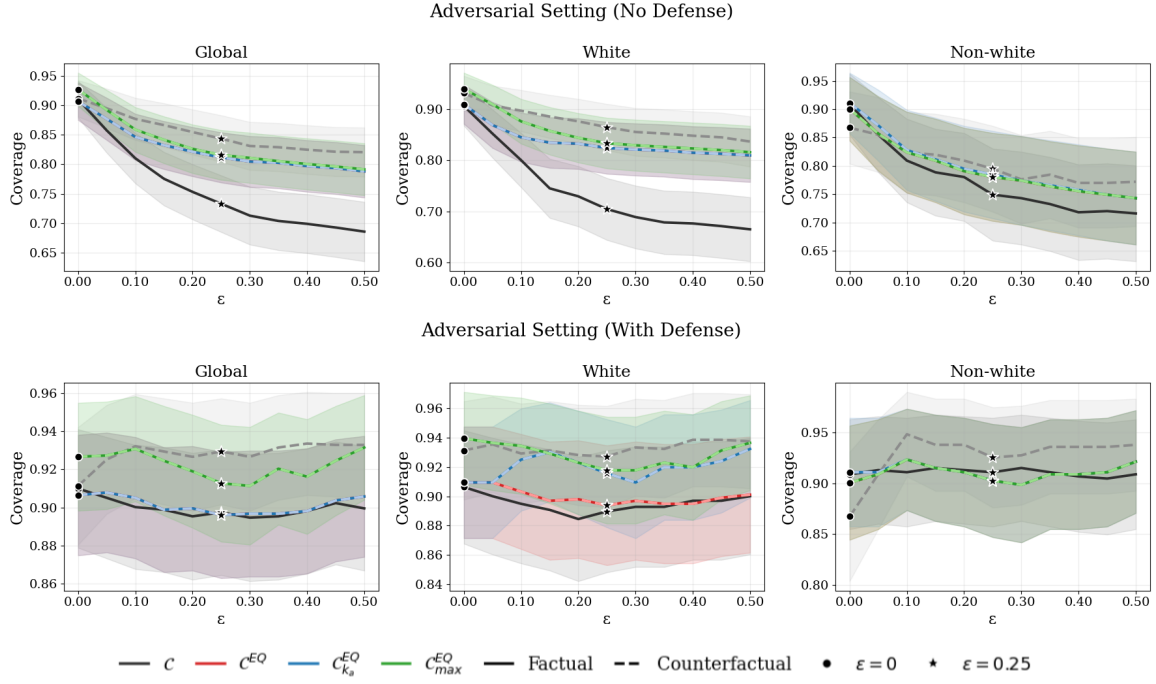


Figure 1: **Coverage under group attacks (A_1) on COMPAS.** Coverage as a function of the attack strength ε , reported globally and per sensitive group. Stars indicate $\varepsilon = 0.25$ (Table 3).

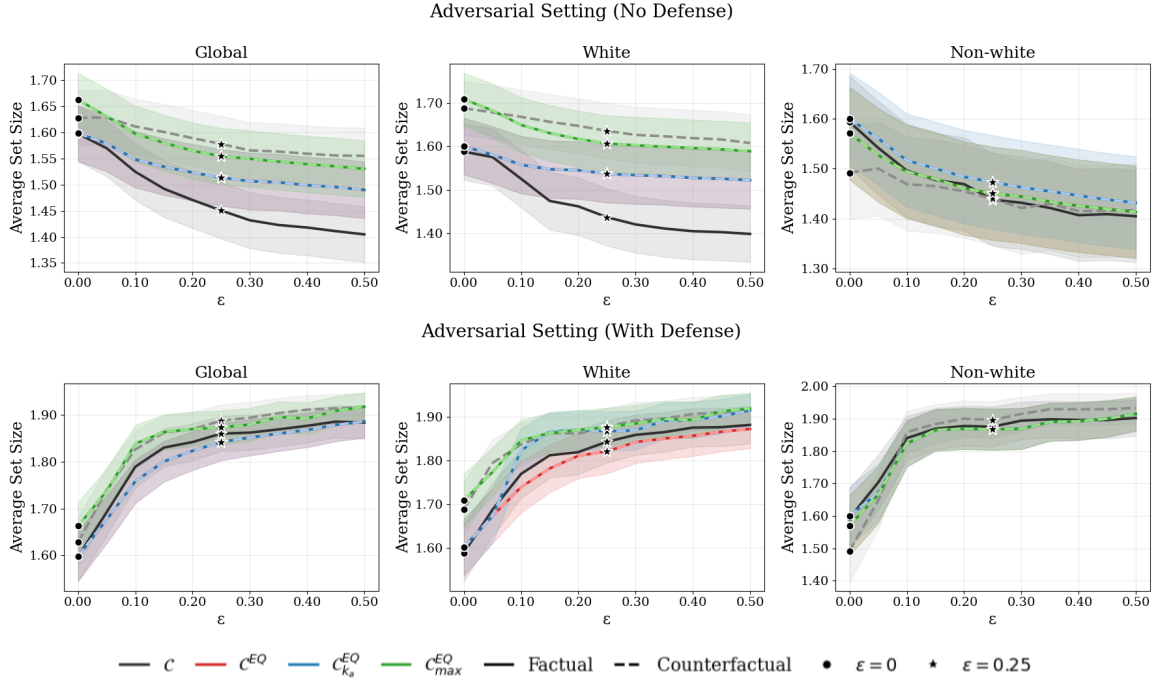


Figure 2: **Average set size under group attacks (A_1) on COMPAS.** Average prediction set size as a function of ε , reported globally and per sensitive group. Stars indicate $\varepsilon = 0.25$ (Table 3).

B.2. Behavior under Group-Targeted Attacks (A_2)

Figures 3 to 6 report the impact of group-targeted attacks (A_2), considering alternatively the attack focused on subgroup $a = 0$ and subgroup $a = 1$.

In stark contrast to the A_1 scenario, these targeted perturbations induce highly asymmetric effects: the attacked group experiences a sharp, monotonic decrease in both coverage and average set size, while the non-attacked group remains largely unaffected. This behavior significantly amplifies fairness violations as ε grows. Our defense effectively counters this worst-case asymmetric degradation by stabilizing the coverage of the targeted group, bringing it back to the nominal threshold, while the average set size moves monotonically toward an upper plateau representing the maximum attainable set size under strong adversarial conditions.

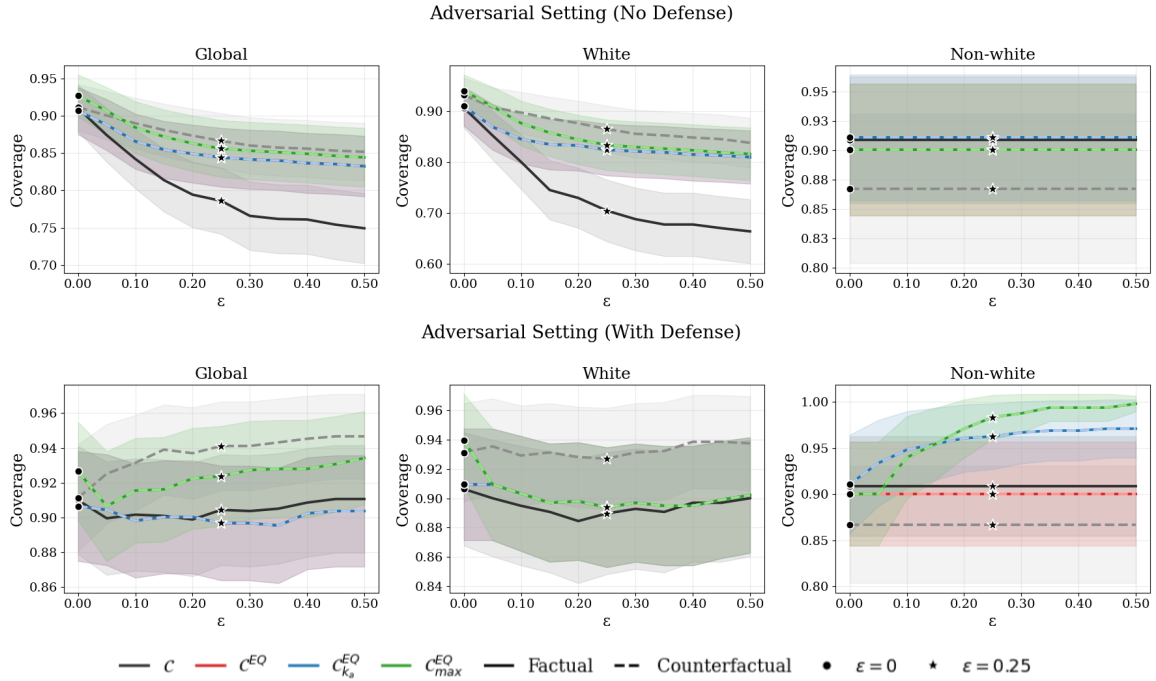


Figure 3: **Coverage under targeted attacks (A_2) on COMPAS (attack on $a = 0$).** Coverage as a function of ε , showing asymmetric degradation across groups. Stars indicate $\varepsilon = 0.25$ (Table 4).

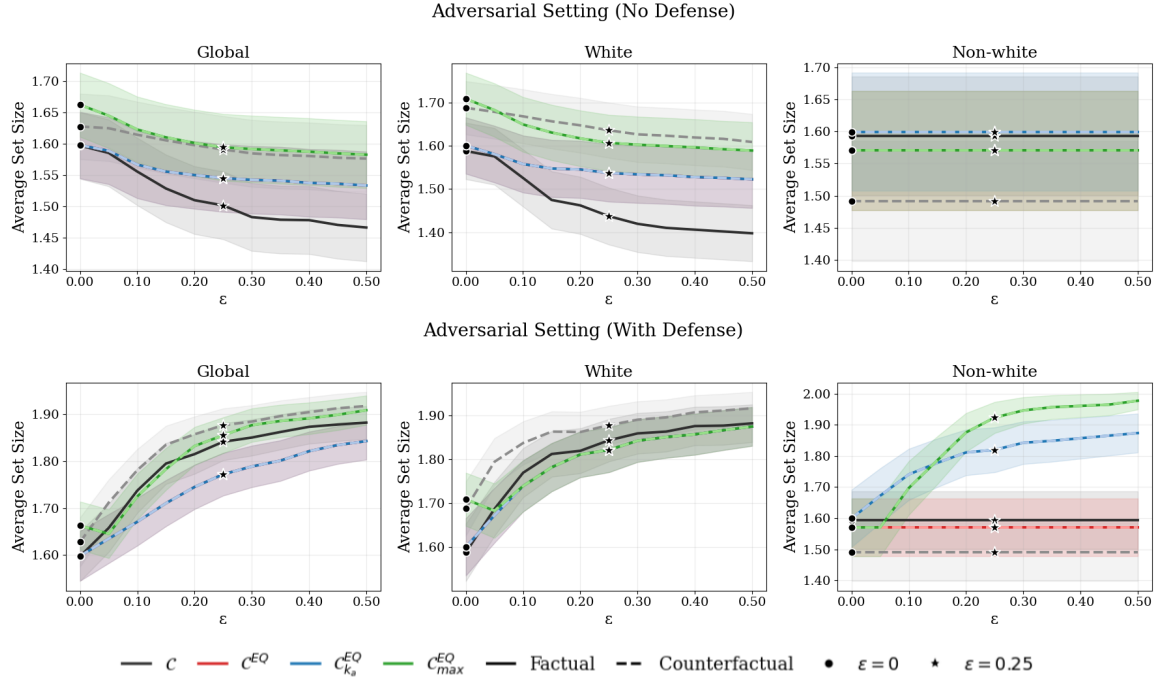


Figure 4: **Average set size under targeted attacks (A_2) on COMPAS (attack on $a = 0$).** Average prediction set size as a function of ϵ , highlighting group-dependent effects. Stars indicate $\epsilon = 0.25$ (Table 4).

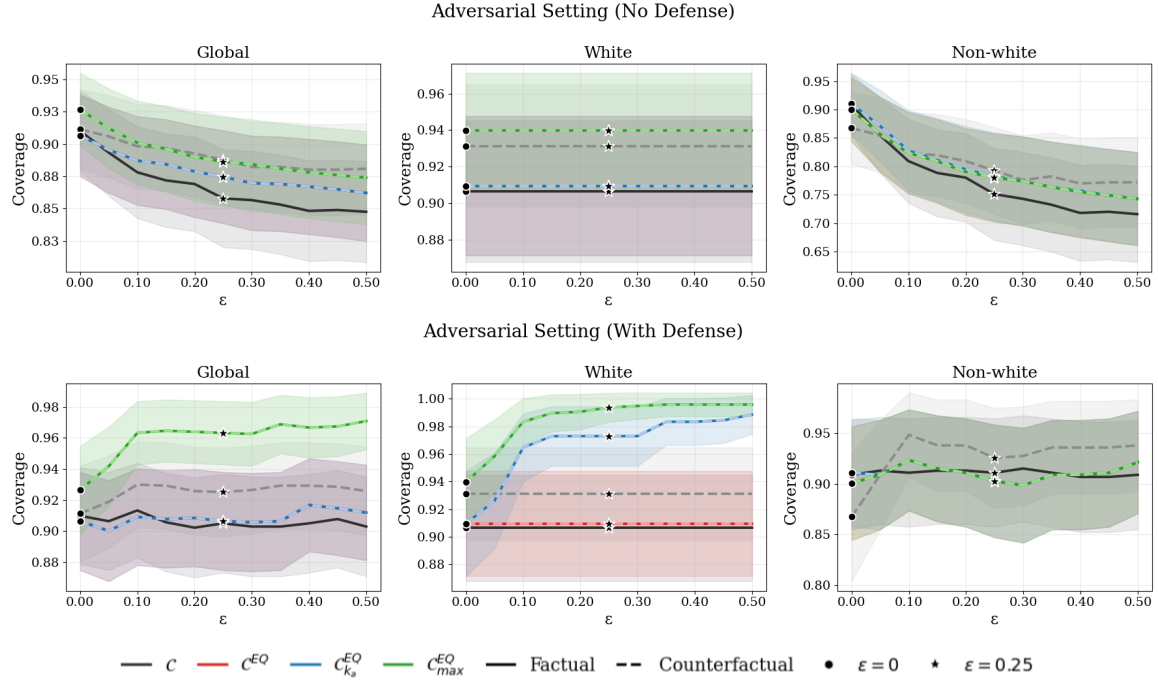


Figure 5: **Coverage under targeted attacks (A_2) on COMPAS (attack on $a = 1$).** Coverage as a function of ϵ , showing asymmetric degradation across groups. Stars indicate $\epsilon = 0.25$ (Table 4).

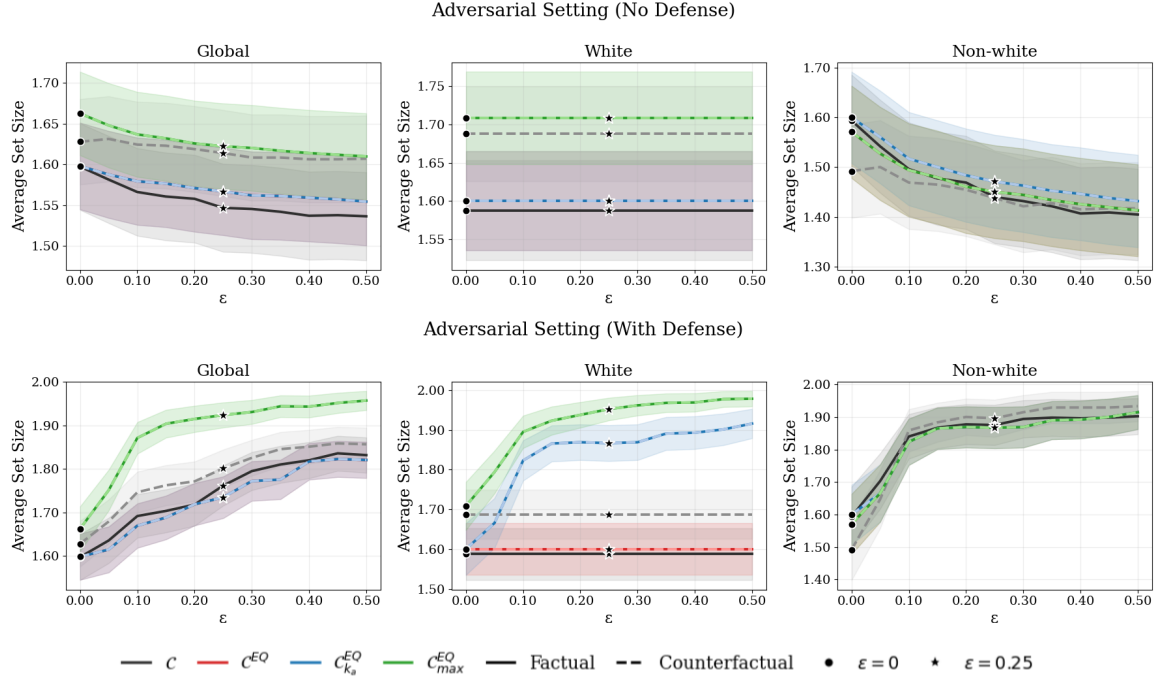


Figure 6: **Average set size under targeted attacks (A_2) on COMPAS (attack on $a = 1$).** Average prediction set size as a function of ϵ , highlighting group-dependent effects. Stars indicate $\epsilon = 0.25$ (Table 4).